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LAB MANUAL

IT Infrastructure

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LAB NO#01

Introduction to IT Infrastructure

Objective:

Introduction to IT infrastructure and its purposes.

IT Infrastructure refers to the complete set of hardware, software, networks, facilities, and services needed to develop, test, deliver, monitor, control, or support IT services within an organization.

IT infrastructure supports communication, data storage, security, and business operations. It includes hardware devices (servers, routers, computers), software (operating systems, databases, cloud platforms), and communication protocols (TCP/IP, HTTP, FTP, SMTP, DNS).

Key Purposes of IT Infrastructure

IT infrastructure is designed to support and manage an organization's IT environment.

Communication

- Enables email, messaging, and video conferencing.
- Connects employees, customers, and partners.

Data Management

- Stores, processes, and retrieves data.
- Ensures backup and disaster recovery.

Security

- Protects systems from cyber threats.
- Controls user access and authentication.

Business Operations Support

- Runs business applications (ERP, CRM, HR systems).
- Supports daily operations.

Scalability & Performance

- Allows system expansion as the business grows.
- Ensures high performance and availability.

Remote Access

- Enables work from home through VPN and cloud systems.

Devices Used in IT Infrastructure

End-User Devices

- Desktop computers
- Laptops
- Tablets
- Smartphones

Networking Devices

- Router
- Switch
- Hub
- Modem
- Access Point
- Firewall appliance

Server & Storage Devices

- Physical servers
- Virtual servers
- NAS (Network Attached Storage)
- SAN (Storage Area Network)
- Backup devices

Security Devices

- Hardware firewalls
- IDS/IPS systems
- Biometric devices

Software Used in IT Infrastructure

- Operating Systems
- Windows Server
- Linux (Ubuntu, CentOS)
- macOS

Database Software

- MySQL
- Oracle
- SQL Server
- PostgreSQL

Network Management Software

- Cisco Packet Tracer
- SolarWinds
- Nagios

Security Software

- Antivirus
- Security information and event management (SIEM) systems
- Encryption tools

Cloud Platforms

- AWS
- Microsoft Azure
- Google Cloud

Business Applications

- ERP systems
- CRM systems
- Email servers

Protocols Used in IT Infrastructure

Protocols are rules that allow devices to communicate.

Network Protocols

- **TCP/IP** – Core telecommunication protocol
- **HTTP/HTTPS** – Web communication / Hypertext Transfer Protocol
- **FTP** – File Transfer Protocol
- **SMTP** – Simple Mail Transfer Protocol (sending email)
- **POP3/IMAP** – Receiving email protocols
- **DNS** – Domain Name System (name resolution)
- **DHCP** – Automatic IP address assignment

Security Protocols

- **SSL/TLS** – Secure communication
- **SSH** – Secure remote login
- **IPsec** – Secure IP communication

Wireless Protocols

- Wi-Fi (802.11)
- Bluetooth

LAB NO#02

Introduction to Cisco Packet Tracer

What is Cisco Packet Tracer?

Cisco Packet Tracer is a powerful network simulation and visualization tool developed by Cisco Systems. It allows students and network professionals to design, build, and troubleshoot virtual networks using a wide variety of simulated hardware, such as routers, switches, and IoT devices, without needing expensive physical equipment.

Originally designed for the Cisco Networking Academy, it is now widely used for CCNA exam preparation and general networking education.

Quick Facts

- **Developer:** Cisco Systems.
- **Platform Support:** Windows, macOS, and Ubuntu Linux
- **Cost:** Free for users who enroll in an introductory course on Cisco Skills For All or NetAcad.
- **Primary Use:** Network simulation, training, and testing configurations before deployment.

Key Simulation Modes

Packet Tracer offers two distinct environments for network analysis:

- **Logical Workspace:** A schematic environment where you drag-and-drop devices and connect them visually to build network topologies.
- **Physical Workspace:** A realistic view that lets you place devices in virtual racks, wiring closets, and even different cities to understand spatial constraints and cabling.
- **Simulation Mode:** A "time-control" feature that allows you to watch data packets travel across the network in slow motion, helping you inspect protocols and analyze step-by-step communication.

Core Features & Capabilities

- **Device Simulation:** Includes a massive library of Cisco routers, switches, wireless access points, firewalls, and servers.
- **Command Line Interface (CLI):** Users can configure devices using the same commands found in real Cisco **IOS** software.
- **IoT & Cybersecurity:** Modern versions include smart home devices, sensors, and industrial equipment to simulate **Internet of Things (IoT)** and security scenarios.
- **Multi-User Support:** Allows multiple students to collaborate on a single network design from different computers.

Why Use It?

- **Risk-Free Learning:** You can’t "break" a virtual router. It’s the perfect place to make mistakes while learning the Command Line Interface (CLI).
- **Certification Prep:** It is the primary tool used for candidates studying for the **CCNA (Cisco Certified Network Associate)**.
- **Visualization:** It turns abstract concepts (like the OSI Model or spanning tree protocols) into visible animations.

Installation & Setup of Cisco Packet Tracer

Objective: Download and set up Cisco Packet Tracer.

Hardware Used:

- MECHREVO Series Laptop
- AMD Ryzen 5 7430U with Radeon Graphics @ 2.30 GHz
- 8 GB RAM
- 238 GB SSD
- AMD Radeon integrated graphics

Windows 11 Pro Education (Version 25H2, 64-bit)

Software Used:

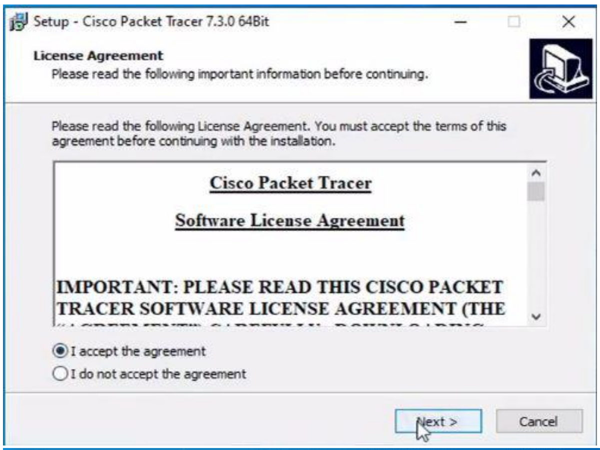
Cisco Packet Tracer

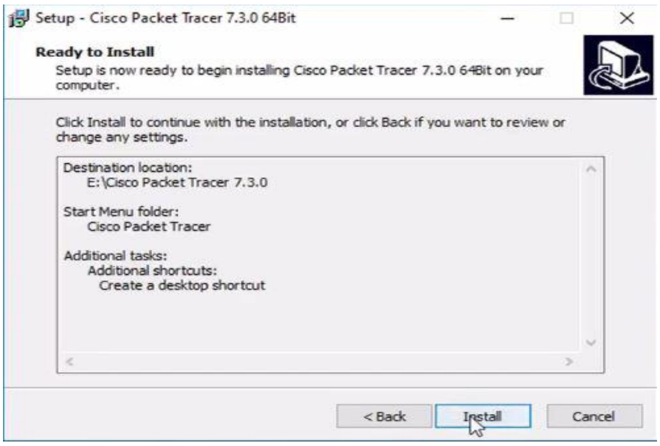
Steps:

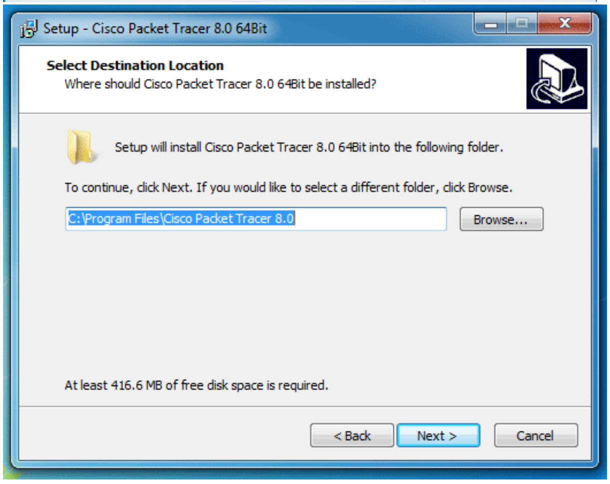
- Go to https://www.netacad.com.
- Sign up or log in to your Cisco NetAcad account.
- Enroll in the “Introduction to Packet Tracer” course.

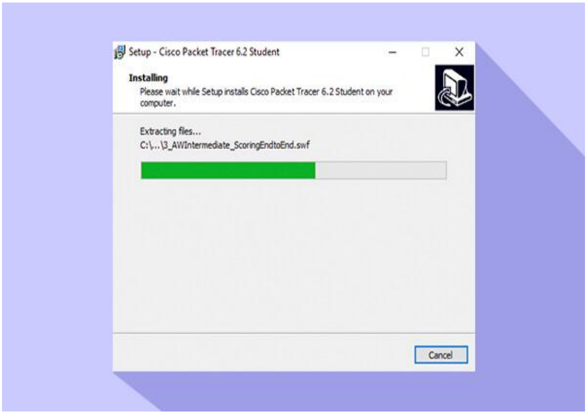
➢ Go to the course resources and download Packet Tracer (choose your operating system).
➢ Run the installer and follow the prompts to install.
➢ Open Packet Tracer and log in with your Cisco account.

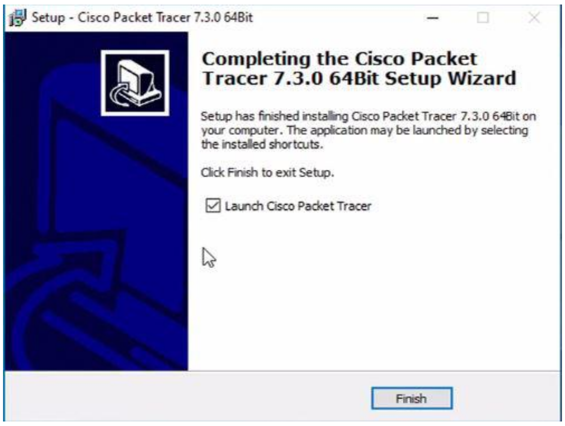
Downloading Process of Cisco Packet Tracer

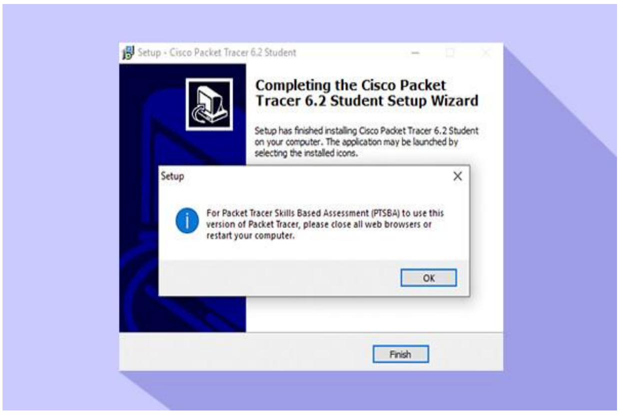




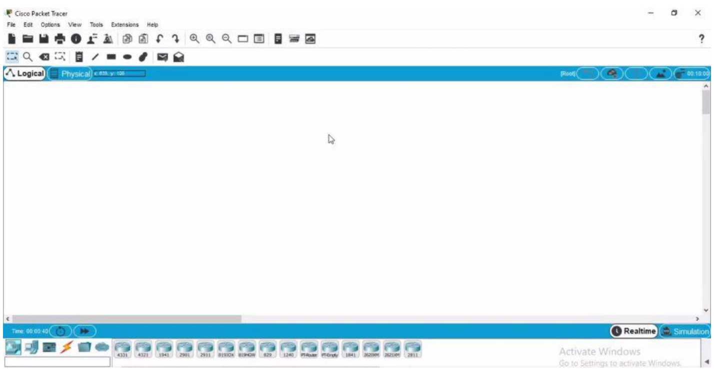








Interface of Cisco Packet Tracer



LAB NO#03

Introduction to Straight Through & Cross wired Cables Tools

Straight-through and cross-wired (crossover) cables are Ethernet cables used to connect network devices. Straight-through cables (T568B to T568B) connect dissimilar devices (e.g., computer to switch), while crossover cables connect similar devices (e.g., switch to switch). They are crafted using CAT5e/CAT6 cables, RJ45 connectors, and a crimping tool.

Straight-Through Cable (Patch Cable)

- **Wiring:** Both ends are wired identical to the T568A or T568B standard (pin 1 connects to pin 1, pin 2 to pin 2).
- **Usage:** Connects *unlike* devices: Computer to Switch, Router to Switch, Computer to Hub.
- **Purpose:** Standard networking cable to connect hosts to networking equipment.

Cross-Wired Cable (Crossover Cable)

- **Wiring:** The transmit (\(TX\)) and receive (\(RX\)) lines are crossed, typically connecting pin 1 to 3 and 2 to 6 on opposite ends.
- **Usage:** Connects *like* devices: Switch to Switch, Computer to Computer, Router to Router.
- **Purpose:** Allows direct communication between similar devices without a switch or router. *Note: Most modern devices use Auto MDI-X to detect this, making crossover cables less common today.*

Essential Cable Assembly Tools

Making your own Ethernet cables requires specific tools:

- **Ethernet Cable:** Usually Category 5e or 6 (UTP).
- **RJ45 Connectors:** Plastic connectors placed on the end.
- **Crimping Tool:** Used to lock the RJ45 connectors onto the cable permanently.
- **Wire Stripper/Cutter:** Used to remove the outer jacket without damaging the inner copper wires.
- **Cable Tester:** A tool used to verify that each pin is connected correctly and the cable works.

Key Differences

Feature	Straight-Through	Crossover
Wiring Standard	Same on both ends (T568B/T568B)	Different on ends (T568A and T568B)
Connections	Unlike (PC to Switch)	Like (PC to PC, Switch to Switch)
Common Use	Daily LAN connections	Direct connection/Legacy
Popularity	Very High	Low (replaced by Auto MDI-X)

Wiring Standards (T568B Example)

1. White Orange
2. Orange
3. White Green
4. Blue
5. White Blue
6. Green
7. White Brown
8. Brown

Creation of Straight Through & Cross wired Cables using Crimping tools

Creating straight-through and crossover Ethernet cables requires Cat5e/Cat6 cable, RJ45 connectors, and a crimping tool. Straight-through cables use the same T568B standard on both ends for connecting different devices (e.g., PC to Switch). Crossover cables use T568A on one end and T568B on the other to connect similar devices (e.g., PC to PC).

Required Tools and Materials

- **Twisted-pair Cable:** Cat 5, 5e, 6, or 7.
- **RJ-45 Connectors:** Standard modular connectors.
- **Crimping Tool:** Used to cut, strip, and attach the RJ-45 connector.
- **Cable Stripper/Scissors:** Optional, for removing the outer jacket.

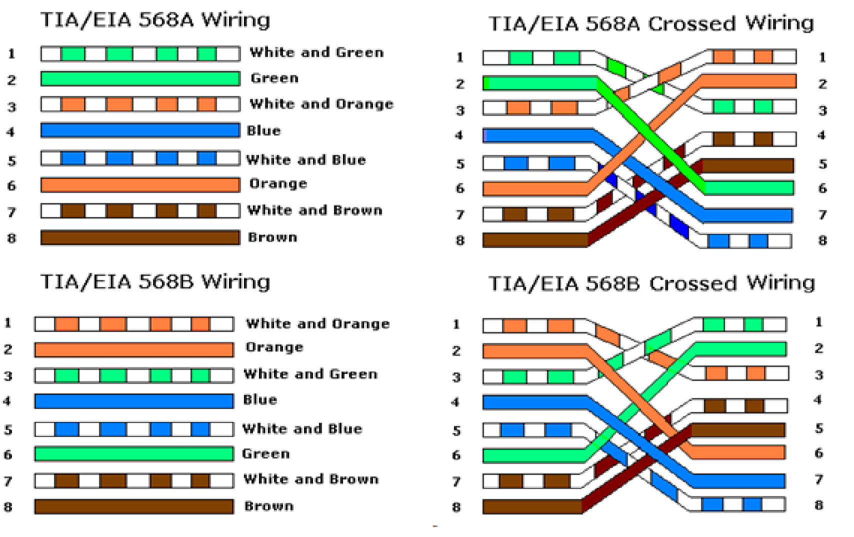


Figure A

Shows the Pin Out of Straight through Cables

Figure B

Shows the Pin Out of Crossover Cables

Step-by-Step Creation Process

1. **Strip the Cable:** Use the crimping tool to strip about 1–1.5 inches of the outer insulation from both ends, taking care not to damage the inner copper wires.
2. **Untwist and Arrange Wires:** Untwist the pairs and arrange them based on the desired standard (A or B).
3. **Trim Wires:** Trim the wires in a straight line, leaving about 0.5 inches exposed from the jacket, to ensure they all reach the end of the connector.
4. **Insert into Connector:** Insert the wires firmly into the RJ-45 connector, ensuring the jacket is inside the connector for, and all wires reach the copper contacts.

5. **Crimp the Connector:** Place the connector into the crimping tool and squeeze firmly to secure the wire and create the connection.
6. **Test the Cable:** Use a cable tester to confirm the connections are accurate.

Wiring Standards (T568A vs T568B)

The T568B standard is more common.

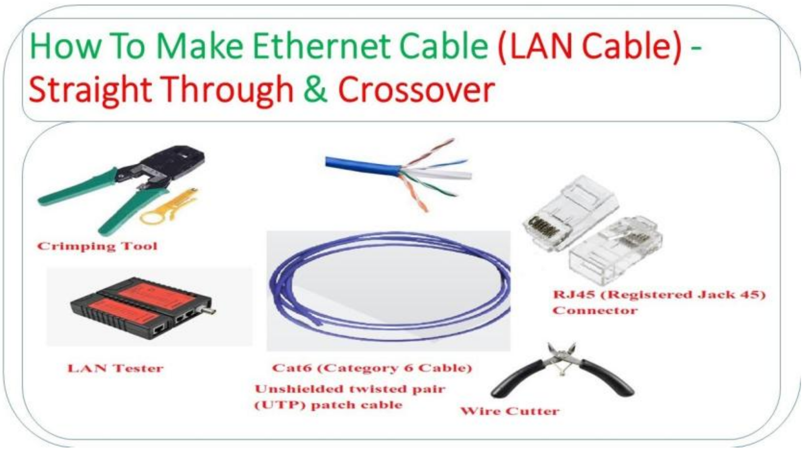
- **T568B:** White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
- **T568A:** White-Green, Green, White-Orange, Blue, White-Blue, Orange, White-Brown, Brown.

Cable Configurations

- **Straight-Through:** Both ends use T568B.
- **Crossover:** One end uses T568A, the other uses T568B.

Important Tips

- Ensure the wire pairs are untwisted as little as possible to prevent signal interference.
- If the wires do not reach the end of the connector, the connection will fail.
- If the cramp fails, cut off the connector and start again with a new one.



LAB NO#4

Experiment NO-1

Objective: Make a simple network

Requirements:

- Router 1x
- PC 2x
- Cables (Straight wire)

Process:

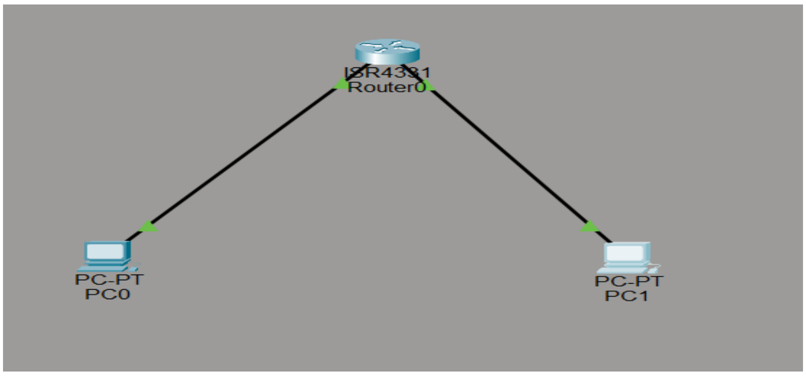
Assign the IP address to both PCs of Class A by going into **Desktop IP Config> Static IP**

IPv4 & subnet Mask

- **PC1:** 192.168.10.2, 255.255.255.255
- **PC2:** 192.168.10.3, 255.255.255.255

Router:

- Go to **physical - > Switch on - > Router.**
- Connects the Wire to **PC1 - > R - > PC2.**



Experiment No.2

Objective:

Configure static Routing using Two Different classes.

Requirements:

- Router 1x (ISR 4331)
- PC 6x (3 for each class)
- Switch 2x (2960-24TT)
- Cables (Straight wire)

Process:

- Divide the pcs into pairs of two groups/classes, each having 3x pcs and 1x switch.
- Design **office A = Class A** Ip address, and **office B = Class B** Ip address**.**
- Now **office A = Class A** Ip address **126.128.0.1** and **office B = Class B** Ip address **191.126.10.1.**
- Both **Classes and Office A, B** both connected with 2x **Switches** via Stright wire to a **router** (working as central hub).

Process for Office A using class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask →Default Gateway.**
- **Pc0:** Ip address of pc0 using class A is **126.128.0.2, 255.0.0.0, 126.128.0.1.**
- **Pc1:** Ip address of pc1 using class A is **126.128.0.3, 255.0.0.0, 126.128.0.1.**
- **Pc2:** Ip address of pc2 using class A is **126.128.0.4, 255.0.0.0, 126.128.0.1.**
- **Switch 1:** The switch is connected to the one side of the **router** assigning the Ip address of class A **126.128.0.1** at **GigaBitEthernet0/0/1.**

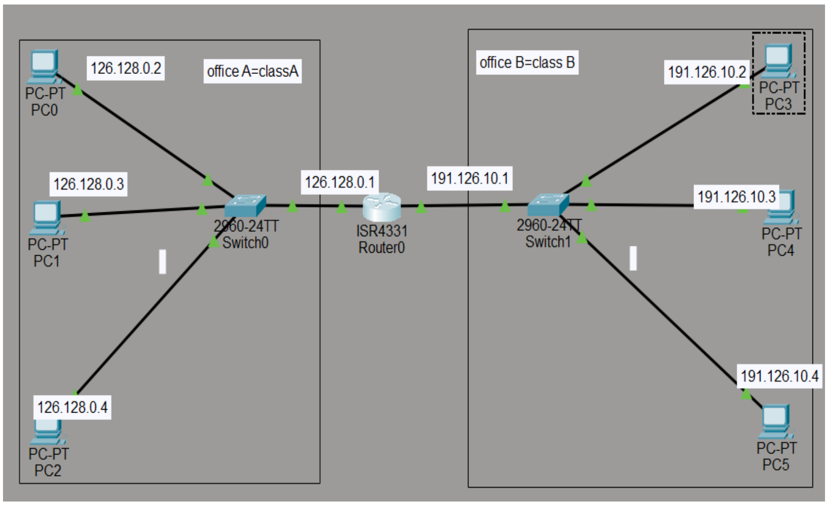
Process for Office B using class B:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc3:** Ip address of pc3 using class B **191.126.0.2, 255.255.0.0, 191.126.10.1.**
- **Pc4:** Ip address of pc4 using class B is **191.126.0.3, 255.255.0.0, 191.126.10.1.**
- **Pc5:** Ip address of pc5 using class B is **191.126.0.4, 255.255.0.0, 191.126.10.1.**

- **Switch 2:** The switch is connected to the second side of the **router** assigning the Ip address of class B **191.126.10.1** at **GigaBitEthernet0/0/0.**

Process of Router Configuration:

- Using **router ISR 4331.**
- From **router** → **config** → **interface** → **GigabitEthernet0/0/1.**
- Connecting **Switch 1** of **126.128.0.1** to **GigaBitEthernet0/0/1** of the router one side of **Class A** from **office A** → Port Status → **On.**
- From **router** → **config** → **interface** → **GigabitEthernet0/0/0.**
- Connecting **Switch 2** of **191.126.10.1** to **GigaBitEthernet0/0/0** of the router one side of **Class B** from **office B** → Port Status → **On.**
- Go to **physical** → **Switch on** → **Router.**



Experiment No.3

Objective:

Configure static routing using Class A with 3 Routers.

Requirements:

- Router 3x (ISR 4331[r0, r1, r2 | R0, R1, R2])
- PC 6x (3 for each class)
- Switch 3x (2960-24TT)
- Straight wire (for pc to switch)
- Serial DTE (for Router to Router)

Process:

- Divide the pcs into pairs of three groups with same class, each having 2x pcs, 1x switch, 1x router.
- Design **office A** = **Class A** Ip address, **office B** = **Class A** Ip address, **office C** = **Class A** Ip address.
- Now, **office A** = **Class A** Ip address **192.168.1.1** and **office B** = **Class A** Ip address **192.168.2.1** **office C** = **Class A** Ip address **192.168.3.1.**
- All PCs of **Office A, B, and C** of **Class A** are connected to 3x **Switches** via Straight wire and Switch are also connected to 3x **routers** (working as data transmitter route/path) via straight wire.
- Then **routers** are connected to each other via **Serial DTE (red colored wire/cable)** cable in order like **r0** → **r1, r1** → **r2.**

Process for Office A using Class A:

- Assign IP Address to PCs by going into **Desktop IP Config, Static IP IPv4 → Subnet Mask -> Default Gateway.**
- **Pc0:** Ip address of pc0 using class A is **192.168.1.2, 255.255.255.0, 192.168.1.1.**
- **Pc1:** Ip address of pc1 using class A is **192.168.1.3, 255.255.255.0, 192.168.1.1.**
- **Switch 1:** The switch is connected to **Fa0/3** side of the **router R0** assigning the Ip address of class A **192.168.1.1, 255.255.255.0** at **GigabitEthernet0/0.**

Process for Office B using Class A:

- Assign IP Address to PCs by going into **Desktop IP Config, Static IP IPv4 → Subnet Mask, and Default Gateway.**
- **Pc2:** Ip address of pc2 using class B **192.168.2.2, 255.255.255.0, 192.168.2.1.**
- **Pc3:** Ip address of pc3 using class B is **192.168.2.3, 255.255.255.0, 192.168.2.1.**
- **Switch 2:** The switch is connected to **Fa0/3** side of the **router R1** assigning the Ip address of class B **192.168.2.1, 255.255.255.0** at **GigaBitEthernet0/0.**

Process for Office C using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc4:** Ip address of pc4 using class A is **192.168.3.2, 255.255.255.0, 192.168.3.1.**
- **Pc5:** Ip address of pc5 using class A is **192.168.3.3, 255.255.255.0, 192.168.3.1.**
- **Switch 3:** The switch is connected to **Fa0/3** side of the **router R3** assigning the Ip address of class A **192.168.3.1, 255.255.255.0** at **GigaBitEthernet0/0.**

Process of Router Configuration R0:

- Using **R0 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1.**
- Connecting **Switch 0** of **192.168.1.1, 255.255.255.0** to **GigaBitEthernet0/0** of the router one side **Class A** from **office A ->** Port Status -> **On.**
- Add **Serial 0/1 of 10.0.0.1, 255.0.0.0** in **R0** router.
- From **router → config → Routing → Static → Network → Mask → Next Hop → Add.**
- At this stage we will add **two routers** in R0.
- 1st Ip of **R1** is **192.168.2.0, 255.255.255.0, 10.0.0.2.**
- 2nd Ip of **R2** is **192.168.3.0, 255.255.255.0, 10.0.0.2.**
- Go to **physical → Switch on → Router.**

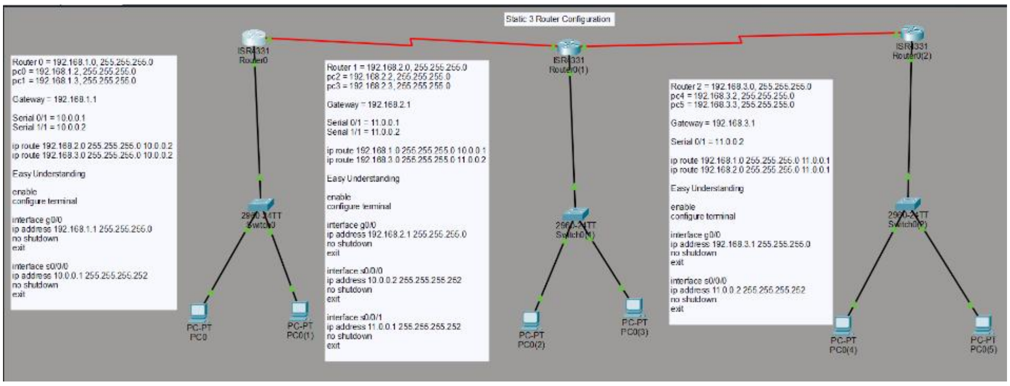
Process of Router Configuration R1:

- Using **R1 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1 → Serial1/1.**
- Connecting **Switch 1** of **192.168.2.2.1, 255.255.255.0** to **GigaBitEthernet0/0** of the router one side of **Class A** from **office B →** Port Status **→ On.**
- Add **Serial 0/1 of 10.0.0.2, 255.0.0.0** in **R1** router.
- Add **Serial 1/1 of 11.0.0.1, 255.0.0.0** in **R1** router.
- From **router → config → Routing → Static → Network → Mask → Next Hop → Add.**
- At this stage we will add **two routers** in R1.

- 1st Ip of **R0** is **192.168.1.0, 255.255.255.0, 10.0.0.1.**
- 2nd Ip of **R2** is **192.168.3.0, 255.255.255.0, 11.0.0.2.**
- Go to **physical → Switch on → Router.**

Process of Router Configuration R2:

- Using **R2 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1.**
- Connecting **Switch 2** of **192.168.3.1, 255.255.255.0** to **GigaBitEthernet0/0** of the router one side of **Class A** from **office C** → Port Status → **On.**
- Add **Serial 0/1 of 11.0.0.2, 255.0.0.0** in **R2** router.
- From **router → config → Routing → Static → Network → Mask → Next Hop → Add.**
- At this stage we will add **two routers** in R2.
- 1st Ip of **R0** is **192.168.1.1.0, 255.255.255.0, 11.0.0.1.**
- 2nd Ip of **R1** is **192.168.2.0, 255.255.255.0, 11.0.0.1.**
- Go to **physical → Switch on → Router.**



LAB NO#05

Experiment No.1

Objective:

Dynamic Router Configure using RIP 3 Routers.

Requirements:

- Router 3x (ISR 4331[r0, r1, r2 | R0, R1, R2])
- PC 6x (3 for each class)
- Switch 3x (2960-24TT)
- Straight wire (for pc to switch)
- Serial DTE (for Router to Router)
- RIP (Router Information Protocol)

Process:

- Divide the pcs into pairs of three groups with same class, each having 2x pcs, 1x switch, 1x router.
- Design **office A = Class A** Ip address, **office B = Class A** Ip address, **office C = Class A** Ip address.
- Now **office A = Class A** Ip address **10.0.0.1** and **office B = Class A** Ip address **20.0.0.1 office C = Class A** Ip address **30.0.0.1.**

- All PCs of **Office A, B, and C** of the **Class A** are connected to 3x **Switches** via Stright wire and Switch are also connected to 3x **routers** (working as data transmitter route/path) via straight wire.
- Then **routers** are connected to each other via **Serial DTE (red colored wire/cable)** cable in order like **r0→r1, r1→r2.**

Process for Office A using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc0:** Ip address of pc0 using class A is **10.0.0.2, 255.0.0.0, 10.0.0.1.**
- **Pc1:** Ip address of pc1 using class A is **10.0.0.3, 255.0.0.0, 10.0.0.1.**
- **Switch 1:** The switch is connected to **Fa0/3** side of the **router R0** assigning the Ip address of class A **10.0.0.1, 255.0.0.0** at **GigabitEthernet0/0.**

Process for Office B using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc2:** Ip address of pc2 using class A **20.0.0.2, 255.0.0.0, 20.0.0.1.**
- **Pc3:** Ip address of pc3 using class A is **20.0.0.3, 255.0.0.0, 20.0.0.1.**
- **Switch 2:** The switch is connected to the **Fa0/3** side of the **router R1** assigning the Ip address of class A **20.0.0.1, 255.0.0.0** at **GigaBitEthernet0/0.**

Process for Office C using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc4:** Ip address of pc4 using class A is **30.0.0.2, 255.0.0.0, 30.0.0.1.**
- **Pc5:** Ip address of pc5 using class A is **30.0.0.3, 255.0.0.0, 30.0.0.1.**
- **Switch 3:** The switch is connected to **Fa0/3** side of the **router R3** assigning the Ip address of class A **30.0.0.1, 255.0.0.0** at **GigaBitEthernet0/0.**

Process of Router Configuration R0:

- Using **R0 router ISR 4331.**
- From **router** → **config** → **interface** → **GigabitEthernet0/0** → **Serial0/1.**
- Connecting **Switch 0** of **10.0.0.1, 255.0.0.0** to **GigaBitEthernet0/0** of the router one side **Class A** from **office A** → Port Status → **On.**
- Add **Serial 0/1 of 40.0.0.1, 255.0.0.0** in **R0** router.
- From **router** → **config** → **Routing** → **RIP** → **Network** → **RIP Routing** → **Add.**
- At this stage we will add **two router rips** in R0.
- 1st **Network Address** of **R0** is **10.0.0.0**
- 2nd **Network Address** of **R0** is **40.0.0.0.**
- Go to **physical** → **Switch on** → **Router.**

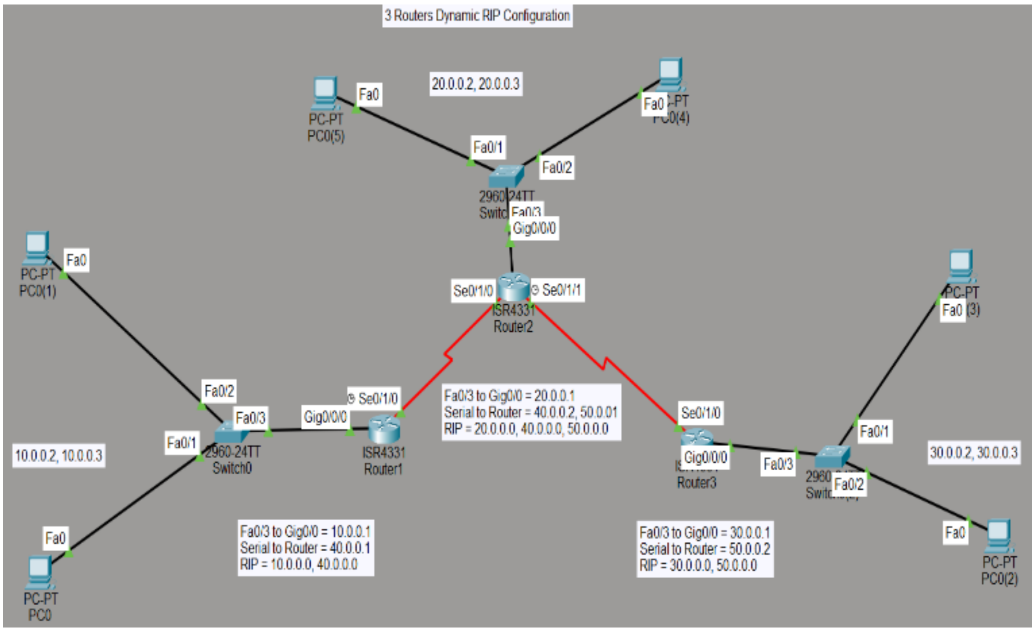
Process of Router Configuration R1:

- Using **R1 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1 →Serial1/1.**
- Connecting **Switch 1** of **20.0.0.1, 255.0.0.0** to **GigaBitEthernet0/0** of the router one side of **Class A** from **office B →** Port Status **→ On.**

- Add **Serial 0/1 of 40.0.0.2, 255.0.0.0** in **R1** router.
- Add **Serial 1/1 of 50.0.0.1, 255.0.0.0** in **R1** router.
- From **router → config → Routing → RIP → Network → RIP Routing → Add.**
- At this stage we will add **three router rips** in R1.
- 1st **Network Address** of **R1** is **20.0.0.0**
- 2nd **Network Address** of **R1** is **40.0.0.0.**
- 3rd **Network Address** of **R1** is **50.0.0.0.**
- Go to **physical → Switch on → Router.**

Process of Router Configuration R2:

- Using **R2 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1.**
- Connecting **Switch 2** of **30.0.0.1, 255.0.0.0** to **GigaBitEthernet0/0** of the router one side of **Class A** from **office C →** Port Status **→ On.**
- Add **Serial 0/1 of 50.0.0.2, 255.0.0.0** in **R2** router.
- From **router → config -> Routing →RIP → Network → RIP Routing → Add.**
- At this stage we will add **two router rips** in R2.
- 1st **Network Address** of **R2** is **30.0.0.0**
- 2nd **Network Address** of **R2** is **50.0.0.0.**
- Go to **physical → Switch on → Router.**



Conclusion

In this experiment, we will see how to configure and set up dynamic rip routing network using same class. This network uses 6x pcs, 3x switches (3x pcs group with 1x switch) each with **Class A** (form a group of 3) connected by 3x routers. The 6x pcs, 3x switches are connected using straight wire and 3x routers relate to each other using **serial DTE** wire and the connection is shown by **green light** at both ends of the switches and routers. The connection is set up from **Office A Switch0** Fa0/3 connected with **router0 Giga0/0/0**, **Office B Switch1** Fa0/3 connected to **router1 Giga0/0/0**. **Office C Switch2** Fa0/3 connected to **router2 Giga0/0/0**. Thus, this creates the simple dynamic rip routing configuration network, which is set up between **office A**, **B and C**, and files can be shared from **Office A pcs** to **Office B pcs** to **Office C pcs** via routers.

Experiment No.2

Objective:

Dynamic RIP Router Configure in Triangular Topology using 3 Routers.

Requirements:

- Router 3x (ISR 4331[r0, r1, r2 | R0, R1, R2])
- PC 6x (3 for each class)
- Switch 3x (2960-24TT)
- Straight wire (for pc to switch)
- Serial DTE (for Router to Router)
- RIP (Router Information Protocol)

Process:

- Divide the pcs into pairs of three groups with same class, each having 2x pcs, 1x switch, 1x router.
- Design **office A = Class A** Ip address, **office B = Class A** Ip address, **office C = Class A** Ip address.
- Now **office A = Class A** Ip address **10.0.0.1** and **office B = Class A** Ip address **20.0.0.1 office C = Class A** Ip address **30.0.0.1.**
- All PCs of **Office A, B, and C** of the **Class A** are connected to 3x **Switches** via Stright wire and Switch are also connected to 3x **routers** (working as data transmitter route/path) via straight wire.
- Then **routers** are connected to each other via **Serial DTE (red colored wire/cable)** cable in order like **r0→r1, r1→r2.**
- Since it is a triangular topology, here files transfer from one group to another group via router configuration through cable order like **r0→r1, r1→r2** or vice versa.

Process for Office A using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask →Default Gateway.**
- **Pc0:** Ip address of pc0 using class A is **10.0.0.2, 255.0.0.0, 10.0.0.1.**
- **Pc1:** Ip address of pc1 using class A is **10.0.0.3, 255.0.0.0, 10.0.0.1.**
- **Switch 1:** The switch is connected to **Fa0/3** side of the **router R0** assigning the Ip address of class A **10.0.0.1, 255.0.0.0** at **GigabitEthernet0/0.**

Process for Office B using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc2:** Ip address of pc2 using class A **20.0.0.2, 255.0.0.0, 20.0.0.1.**
- **Pc3:** Ip address of pc3 using class A is **20.0.0.3, 255.0.0.0, 20.0.0.1.**
- **Switch 2:** The switch is connected to the **Fa0/3** side of the **router R1** assigning the Ip address of class A **20.0.0.1, 255.0.0.0** at **GigaBitEthernet0/0.**

Process for Office C using Class A:

- Assign Ip Address to pcs by going into **Desktop → Ip Config → Static Ip → IPv4 → Subnet Mask → Default Gateway.**
- **Pc4:** Ip address of pc4 using class A is **30.0.0.2, 255.0.0.0, 30.0.0.1.**
- **Pc5:** Ip address of pc5 using class A is **30.0.0.3, 255.0.0.0, 30.0.0.1.**

- **Switch 3:** The switch is connected to **Fa0/3** side of the **router R3** assigning the Ip address of class A **30.0.0.1, 255.0.0.0** at **GigaBitEthernet0/0.**

Process of Router Configuration R0:

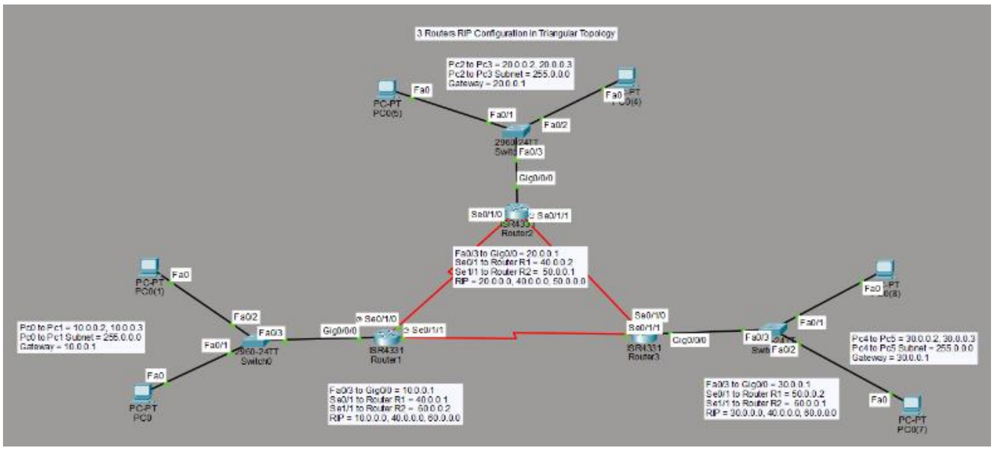
- Using **R0 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1 → Serial1/1.**
- Connecting **Switch 0** of **10.0.0.1, 255.0.0.0** to **GigaBitEthernet0/0** of the router one side **Class A** from **office A →** Port Status **→ On.**
- Add **Serial 0/1 of 40.0.0.1, 255.0.0.0** in **R0** router.
- Add **Serial 1/1 of 60.0.0.2, 255.0.0.0** in **R0** router.
- From **router → config → Routing → RIP → Network → RIP Routing → Add.**
- At this stage we will add **three router rips** in R0.
- 1st **Network Address** of **R0** is **10.0.0.0**
- 2nd **Network Address** of **R0** is **40.0.0.0.**
- 3rd **Network Address** of **R0** is **60.0.0.0.**
- Go to **physical → Switch on → Router.**

Process of Router Configuration R1:

- Using **R1 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 →Serial0/1 → Serial1/1.**
- Connecting **Switch 1** of **20.0.0.1, 255.0.0.0** to **GigaBitEthernet0/0** of the router one side of **Class A** from **office B →**Port Status **→ On.**
- Add **Serial 0/1 of 40.0.0.2, 255.0.0.0** in **R1** router.
- Add **Serial 1/1 of 50.0.0.1, 255.0.0.0** in **R1** router.
- From **router → config → Routing → RIP → Network → RIP Routing → Add.**
- At this stage we will add **three router rips** in R1.
- 1st **Network Address** of **R1** is **20.0.0.0**
- 2nd **Network Address** of **R1** is **40.0.0.0.**
- 3rd **Network Address** of **R1** is **50.0.0.0.**
- Go to **physical → Switch on → Router.**

Process of Router Configuration R2:

- Using **R2 router ISR 4331.**
- From **router → config → interface → GigabitEthernet0/0 → Serial0/1 → Serial1/1.**
- Connecting **Switch 2** of **30.0.0.1, 255.0.0.0** to **GigaBitEthernet0/0** of the router one side of **Class A** from **office C →** Port Status **→ On.**
- Add **Serial 0/1 of 50.0.0.2, 255.0.0.0** in **R2** router.
- Add **Serial 1/1 of 60.0.0.1, 255.0.0.0** in **R2** router.
- From **router → config → Routing → RIP → Network → RIP Routing → Add.**
- At this stage we will add **three router rips** in R2.
- 1st **Network Address** of **R2** is **30.0.0.0**
- 2nd **Network Address** of **R2** is **50.0.0.0.**
- 3rd **Network Address** of **R2** is **60.0.0.0.**
- Go to **physical → Switch on → Router.**



Conclusion:

In this experiment, we will see how to configure and set up triangular topology dynamic rip routing network using same class. This network uses 6x pcs, 3x switches (3x pcs group with 1x switch) each with **Class A** (form a group of 3) connected by 3x routers. The 6x pcs, 3x switches are connected using straight wire and 3x routers relate to each other using **serial DTE** wire and the connections are shown by **green light** at both ends of the switches and routers. The connection is set up from **Office A Switch0** Fa0/3 connected with **router0 Giga0/0/0**, **Office B Switch1** Fa0/3 connected to **router1 Giga0/0/0**. **Office C Switch2** Fa0/3 connected to **router2 Giga0/0/0**. Thus, this creates a simple dynamic rip routing configuration network, since it is triangular which means we can share files from **Office A pcs** to **Office B pcs** to **Office C pcs** via routers.

Experiment No.3

Objective:

Study of Repeater, Uses of Repeater and Connection a simple Repeater.

What is a Repeater?

A repeater is a network device operating at the physical layer (Layer 1) that receives a weakened or distorted signal, regenerates it, and retransmits it to extend network range. It is used to overcome signal attenuation (loss of strength) over long distances or through obstructions, ensuring reliable data transmission without noise.

Key Reasons for Using a Repeater:

- **Extend Network Distance:** It increases the maximum range of LAN cables or wireless signals by retransmitting them, enabling connectivity across large buildings or wide areas.
- **Signal Regeneration/Amplification:** It strengthens signals that have weakened over long distances (attenuation).
- **Reduce Signal Noise:** Unlike simple amplifiers that boost noise along with the signal, repeaters often clean up the signal before transmitting, ensuring data integrity.
- **Overcome Obstacles:** They are used in communication systems to boost signals, helping them overcome physical obstructions.
- **Cost-Effective Expansion:** They provide a budget-friendly solution for extending network coverage in homes and businesses.

Common Types of Repeaters:

- **WiFi Repeaters:** Extend the coverage area of a wireless network.

- **<u>Ethernet Repeaters</u>:** Connect two or more cable segments.
- **<u>Optical Repeaters</u>:** Used to boost light signals in fiber optic cables.
- **<u>Radio/Cellular Repeaters</u>:** Used in telecommunications to extend the range of radio signals.

Configure a simple Repeater Connection using Class C

Requirements:

- PC 6x (3 for each class)
- Switch 2x (2960-24TT)
- Straight wire (for pc to switch)
- 1x Repeater.

Process:

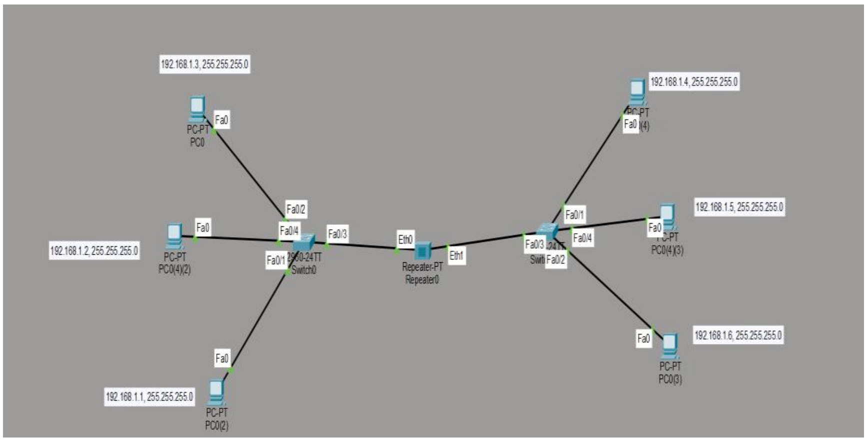
- Divide the pcs into pairs of two groups with same class, each having 3x pcs, 1x switch, 1x repeater (at the center).
- Design **office A = Class C** Ip address, **office B = Class C** Ip address
- Now **office A = Class C** pcs Ip address **192.168.1.1 to 192.168.1.3** and **office B = Class C** pcs Ip address **192.168.1.4 to 192.168.1.6.**
- All PCs of **Office A, B** of the **Class C** are connected to 2x **Switches** via Stright wire and Switch are also connected to **repeater** (working as signal transmitter hub) via straight wire.

Process for Office A Switch:

- Here we all are using a 1x **repeater** connected with **switch**.
- We have 6 pcs (pair of 3) in 2 groups and ip assigned of **Class C.**
- **Pc0** ip address is **192.168.1.1, 255.255.255.0.**
- **Pc1** ip address is **192.168.1.2, 255.255.255.0.**
- **Pc2** ip address is **192.168.1.3, 255.255.255.0.**
- All 3 pcs are connected to **Switch.**
- The **switch** is then connected to **Repeater.**

Process for Office B Switch:

- Here we all are using a 1x **repeater** connected with **switch**.
- We have 6 pcs (pair of 3) in 2 groups and ip assigned of **Class C.**
- **Pc3** ip address is **192.168.1.4, 255.255.255.0.**
- **Pc4** ip address is **192.168.1.5, 255.255.255.0.**
- **Pc5** ip address is **192.168.1.6, 255.255.255.0.**
- All 3 pcs are connected to **Switch.**
- The **switch** is then connected to **Repeater.**



Conclusion:

In this experiment, we observe and learn how to set up the repeater connection in cisco packet tracer. Also learned following things:

- **Successful Full Connectivity:** In the current state shown in the image, all PCs can communicate with each other perfectly. Because they share the same subnet and are connected by Layer 1 (Repeater) and Layer 2 (Switches) devices, the "signals" and "data" flow without restriction from 192.168.1.1 all the way to 192.168.1.6.
- **The Repeater's Role:** The Repeater is extending the "Collision Domain." It is taking the electrical signals from Switch0 and regenerating them for Switch1. It does not look at IP addresses; it simply ensures the signal stays strong enough to reach the other side.

Impact of Changing IP Classes/Masks :

- **If you change a PC to Class A (e.g., 10.0.0.1):** The Repeater will still pass the signal, but the PC will be logically isolated. It will be "shouting" into a room where everyone else is speaking a different language.
- **If you change the Mask to 255.0.0.0:** As we discussed, this creates a "one-way street" or a mismatch. The PC with the wide mask might try to talk to others, but the other PCs (with the 255.255.255.0 mask) will ignore it because they think the sender is on a different network.

Final Verdict

Your diagram represents a single logical Local Area Network (LAN) spread across two physical segments.

- **For Data to flow:** Every PC must have an IP in the same range (like 192.168.1.x) and the same Subnet Mask.
- **The Hardware Limit:** Because you are using a Repeater and Switches (not a Router), you cannot have two different IP classes communicating here. To use different IP classes, you would need to replace that Repeater with a Router.

LAB NO#06

Introduction to Type 2 Hypervisor

A Type 2 hypervisor, also known as a hosted hypervisor, is software that allows you to run multiple operating systems (virtual machines) on top of an existing operating system.

Unlike a Type 1 hypervisor (which runs directly on computer hardware), a Type 2 hypervisor sits on top of a host OS just like any other application (e.g., a web browser or media player).

How It Works

The architecture of a Type 2 hypervisor follows a specific hierarchy:

Hardware – Your physical PC or laptop.

Host Operating System – The primary OS installed on the hardware (Windows, macOS, or Linux).

Type 2 Hypervisor – The software (e.g., VirtualBox, VMware Workstation).

Guest Operating Systems – The virtual machines (VMs) running inside the hypervisor.

Key Characteristics

Ease of Use – Since they run as applications, Type 2 hypervisors are very easy to install and manage. You do not need to wipe your computer or use a dedicated server to get started.

Resource Sharing – The hypervisor requests resources (CPU, RAM, storage) from the host OS. Because the host OS also uses these resources for its own tasks, Type 2 hypervisors usually have slightly more overhead (latency) than Type 1 hypervisors.

Portability – You can easily move a virtual machine file from one computer to another, provided both have the hypervisor software installed.

Popular Examples

Oracle VM VirtualBox – A popular, open-source hypervisor used frequently in labs and for testing different OS environments.

VMware Workstation Pro / Player – A robust industry standard for Windows and Linux users.

VMware Fusion – The macOS-specific version for running Windows or Linux on a Mac.

Parallels Desktop – Optimized for running Windows on Mac hardware.

Use Cases

Testing and Development – Developers use Type 2 hypervisors to test code across different operating systems (e.g., testing a Linux application while on a Windows laptop).

Learning & Labs – As with Cisco Packet Tracer, Type 2 hypervisors are perfect for students who want to build a lab of virtual routers or servers without buying extra hardware.

Running Legacy Applications – If you have an old program that only works on Windows XP, you can run an XP virtual machine on your modern Windows 11 machine.

Installation of a Type 2 Hypervisor

The installation process for a Type 2 hypervisor is very similar to installing any standard desktop application. This guide focuses on Oracle VM VirtualBox, the most popular free, open-source option for students.

Step 1: Check System Requirements

Before installing, ensure your physical computer (the host) has:

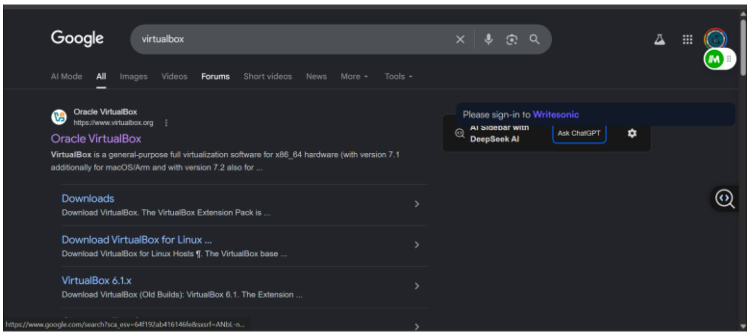
Virtualization Enabled – You must enable Intel VT-x or AMD-V in your computer’s BIOS/UEFI settings. Without this, the hypervisor may fail to launch 64-bit virtual machines.

RAM – At least 8 GB is recommended (the host OS needs some, and the guest VM needs its own share).

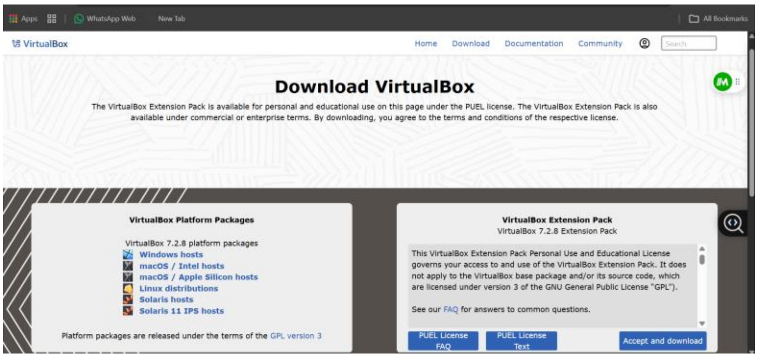
Storage – Enough free space for virtual hard disks (usually 20–50 GB per VM).

Step 2: Download the Installer

- Go to the official website (e.g., virtualbox.org).



- Select the host platform that matches your current OS (Windows, macOS, or Linux).



- Download the Extension Pack (optional but highly recommended) to enable support for USB 3.0 and remote desktop features.

Step 3: Run the Installation Wizard

Run the downloaded .exe (Windows) or .dmg (macOS) file.

Custom Setup – Usually, you can leave the default components selected. VirtualBox will install networking drivers. Note: Your internet connection might briefly drop during this step while virtual network adapters are created.

Click Install and follow the prompts until finished.

Step 4: Set Up Your First Virtual Machine (VM)

Once the hypervisor is open, follow these steps to create a guest:

Click New – Give your VM a name (e.g., “Ubuntu Linux”).

Allocate RAM – Choose how much of your physical RAM to give the VM. Tip: Stay in the “green” zone to avoid starving your host computer of memory.

Create a Virtual Hard Disk – The VDI (VirtualBox Disk Image) format is usually best. Choose Dynamically Allocated so it only takes up space on your physical drive as you add files to the VM.

Mount the ISO – In the VM settings under Storage, click the empty CD icon and select the ISO file (the installer image) of the operating system you want to run (e.g., a Windows 10 ISO or Linux Mint ISO).

Step 5: Start the VM and Install the Guest OS

- Select your VM from the list and click the green Start arrow.
- The VM will boot from the ISO file. You will then go through the OS installation process just as if you were using a brand-new physical computer.

LAB NO#07

Introduction to Kali/Ubuntu/Mint

Linux Mint is a beginner-friendly, open-source operating system based on Ubuntu and Debian, designed to be modern, elegant, and comfortable to use out of the box. It is widely considered an excellent, stable alternative to Windows or macOS, featuring a familiar interface and strong software support without requiring extensive maintenance.

The Three "Flavors" of Mint

When you go download Mint, you'll notice three different versions (Desktop Environments). Choosing the right one depends on your computer's power:

1. **Cinnamon (The Flagship):** The most modern and full-featured version. It has all the visual "eye candy" (shadows, transitions) and is the most popular.
2. **MATE:** A bit more "classic" and uses fewer resources. It's based on how Linux desktops looked about 10 years ago simple and fast.
3. **XFCE:** The "lightweight" champion. If you are running your Type 2 Hypervisor on an older laptop with limited RAM, **this is the one to pick.**

Key Features of Linux Mint

- **User-Friendly Interface:** The default desktop environment, **Cinnamon**, provides a traditional layout with a menu, taskbar, and desktop icons similar to Windows.
- **Out-of-the-Box Functionality:** Includes media codes, drivers, and pre-installed software (browsers, office suites, media players) for immediate use.
- **Software Manager:** Features an easy-to-use Software Manager (app store) to download thousands of applications like Steam, Spotify, and LibreOffice.
- **Stability & Security:** Built on a "conservative" update model, meaning it prioritizes system stability. It is safe from most malware and does not require antivirus software.
- **Low Resource Usage:** Excellent for rejuvenating older hardware.

Key Technical Specs

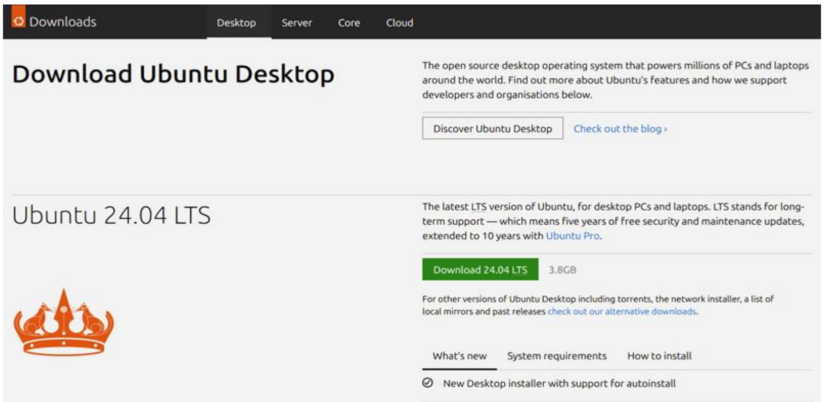
If you are setting this up in **VirtualBox** or **VMware**, here are the recommended settings for a smooth experience:

- **RAM:** Allocate **4GB** (2GB is the minimum, but it will be sluggish).
- **Storage: 20GB** of virtual disk space.
- **Graphics:** Ensure "3D Acceleration" is enabled in your VM settings if you choose the **Cinnamon** edition.

Installation of Kali/Ubuntu/Mint

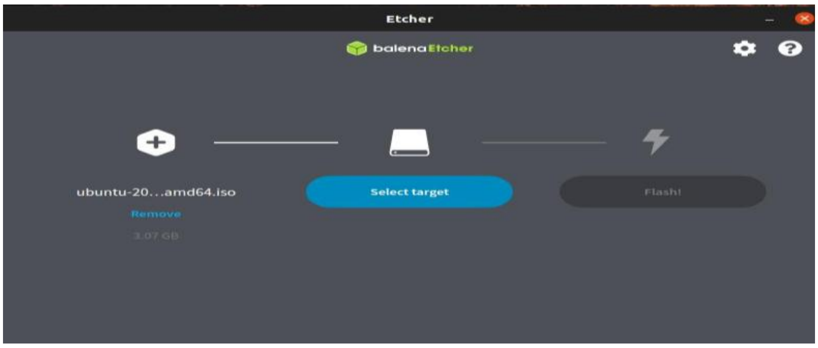
Download the Ubuntu ISO:

Visit the [Ubuntu download page] and download the latest Ubuntu Desktop ISO file.



Create a Bootable USB Drive:

- Use Balen Etcher to create a bootable USB stick.
- Insert a USB drive (at least 12GB) into your computer.
- Open Balen Etcher, select the downloaded Ubuntu ISO, choose your USB flash drive, and click 'Flash!' to create the bootable USB.

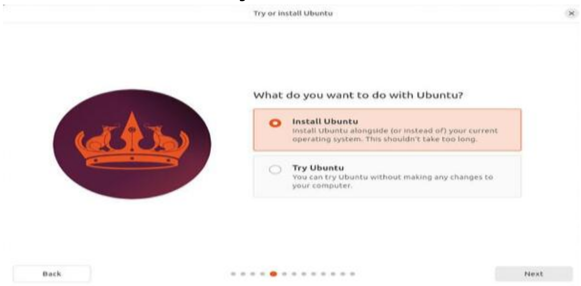


Boot your computer from the USB

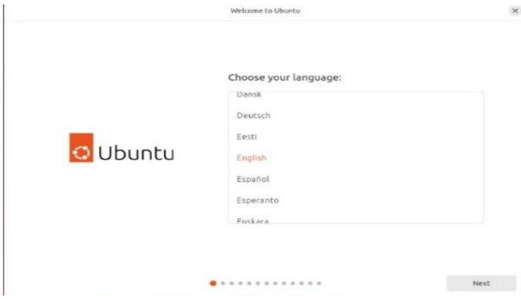
- Insert the bootable USB drive into your computer.
- Restart the computer and access the boot menu (commonly by pressing F12, Esc, F2, or Del during startup).
- Select the USB drive from the boot options to start the Ubuntu installer.

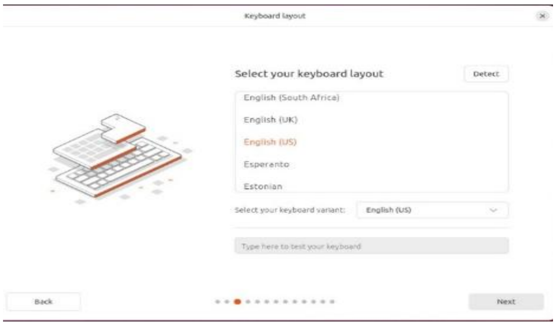
Start the Ubuntu Installation:

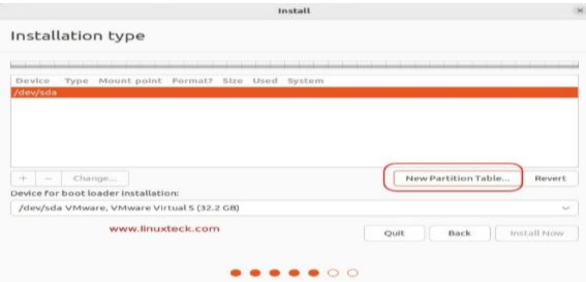
- Once Ubuntu boots, select Try Ubuntu to ensure hardware compatibility.
- When ready, double-click the Install Ubuntu icon on the desktop to begin the installation.



- Follow the Ubuntu installer: choose language, keyboard layout, installation type (e.g., “Install alongside existing OS” or “Erase disk”).







If manual partitioning, create:

- Root partition / (ext4, ≥20 GB)
- Swap partition (size = RAM, optional)
- Home partition /home (remaining space)
- Complete installation, remove USB, and reboot.

Begin Installation

- Review your partition settings and click Install Now.
- Confirm any prompts about writing changes to the disk.

Complete Installation

- Allow the installation process to finish.
- Once done, remove the USB drive when prompted and click Restart Now.

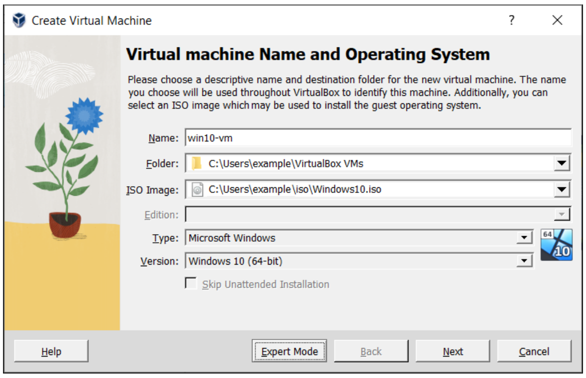
Boot into Ubuntu or Windows:

- Upon rebooting, you'll see the GRUB boot menu, allowing you to select either Ubuntu or Windows to boot into.

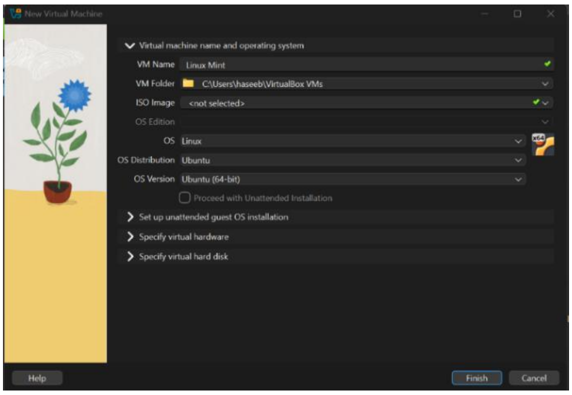
Create a VM in VirtualBox and install a Linux distribution (e.g., Linux Mint).

Step-by-step:

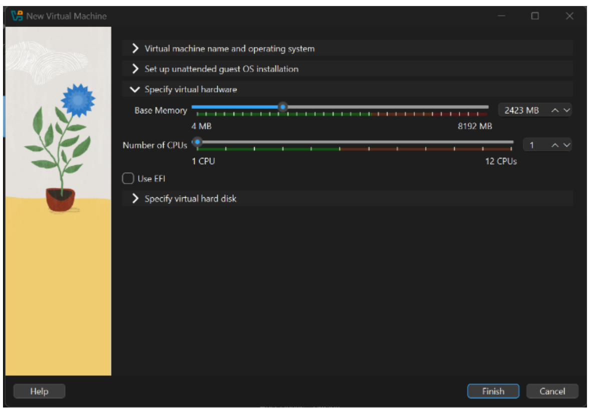
- Open VirtualBox and click New.



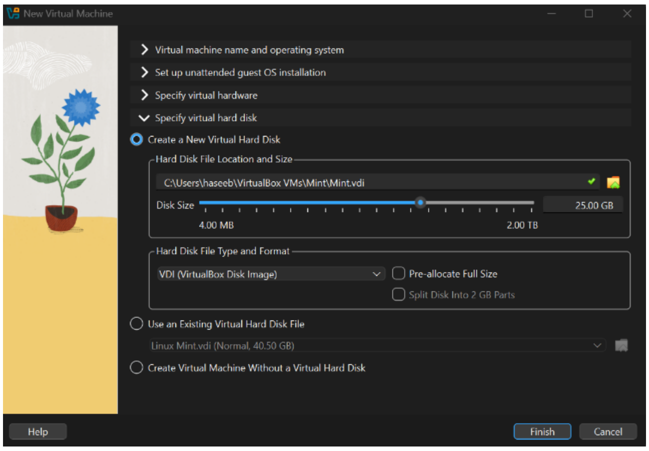
- Name your VM (e.g., “Linux Mint”), select Type: Linux, Version: Ubuntu (64-bit).



- Assign memory (RAM): at least 2 GB, but 4 GB recommended.

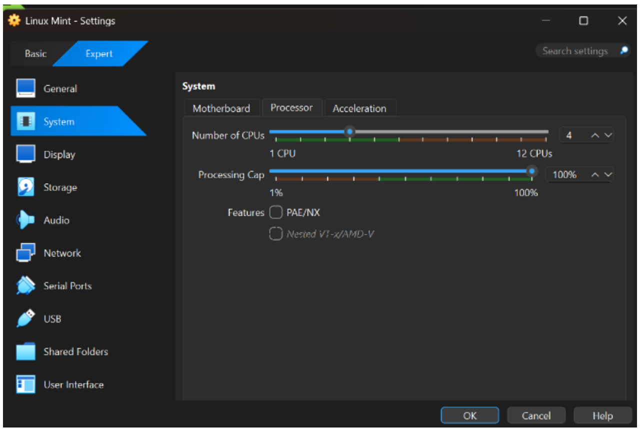


- Create a virtual hard disk: choose VDI (VirtualBox Disk Image), dynamically allocated, size 20-30 GB.

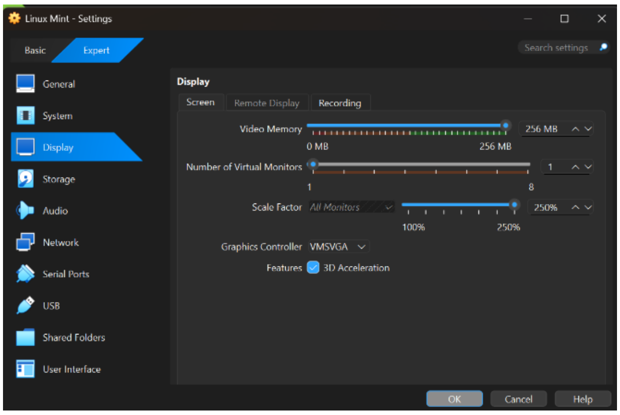


After creation, go to Settings:

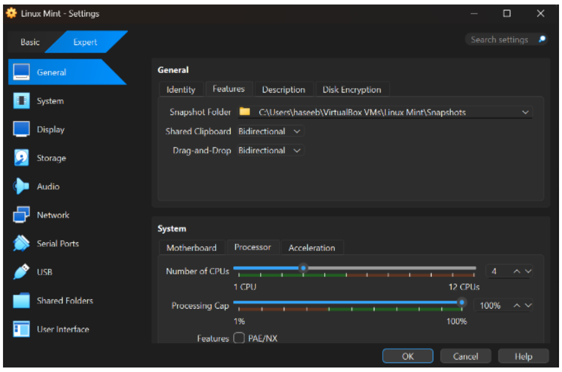
- System → Processor → allocate 4 CPU cores.



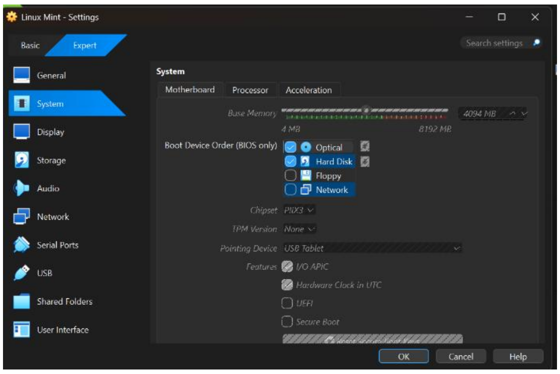
- Display → enable 3D Acceleration, set Video Memory to 256 MB.



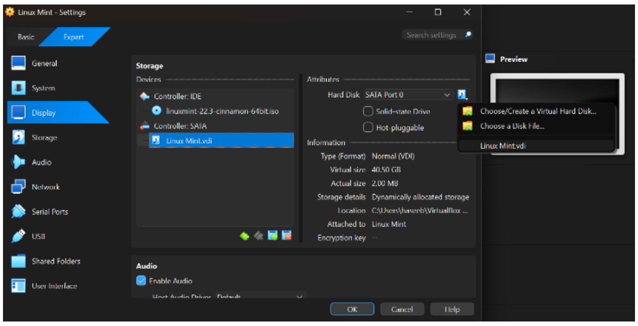
- In General, go to Feature, Shared Clipboard & Drag and Drop → both set to Bidirectional.



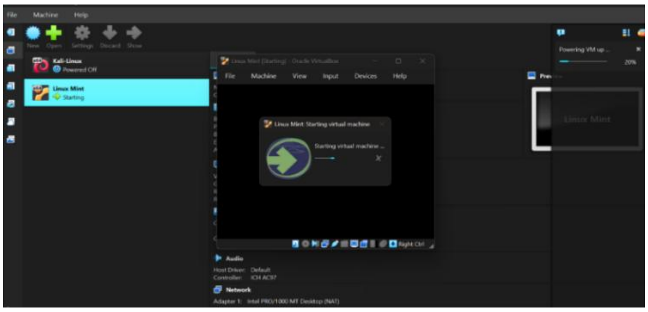
- In System settings, disable the Floppy option and go to the Processor.



- In Storage, load the Linux Mint ISO into the optical drive.



- Start the VM and follow the installer's prompts (language, time zone, user account).



LAB NO#08

Create a Snapshot & Clone of Virtual Machine

Creating a Snapshot in VirtualBox

Snapshot in VirtualBox (Most Common)

If your Kali is running in **Oracle VM VirtualBox**:

Steps:

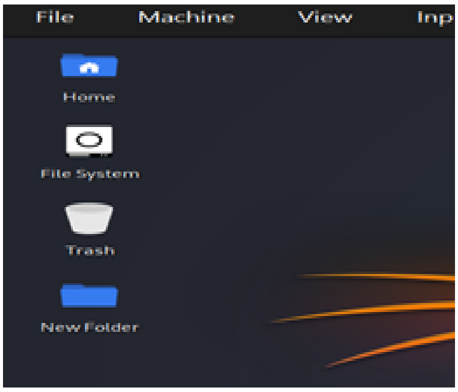
1. Open VirtualBox
2. Select your **Kali Linux VM**
3. Click on **Snapshots** tab
4. Click **Take Snapshot** (or use Host + T)
5. Enter:

- Name (e.g., *Before Update*)
- Description (optional)

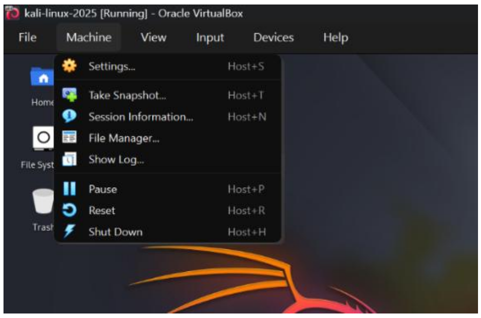
6. Click **OK**

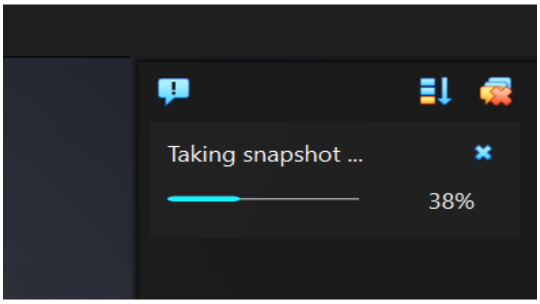
Done! Your snapshot is created

Screen Shot after creating a folder



Selecting the Machine Menu for screen shot

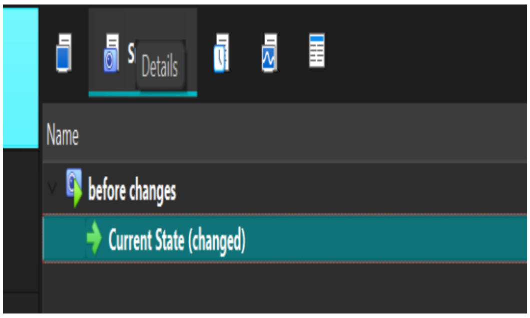


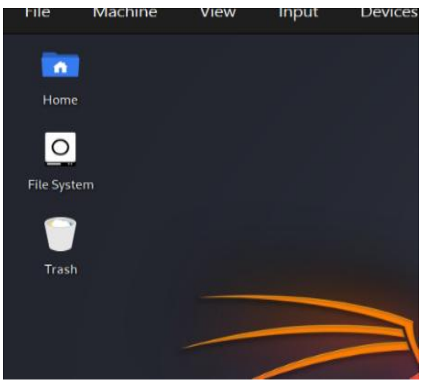


Process of Snapshot

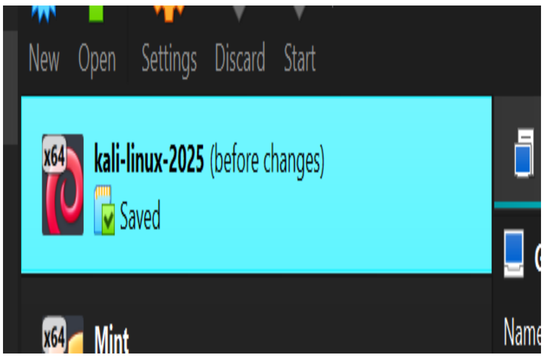
Vs

Completed Process





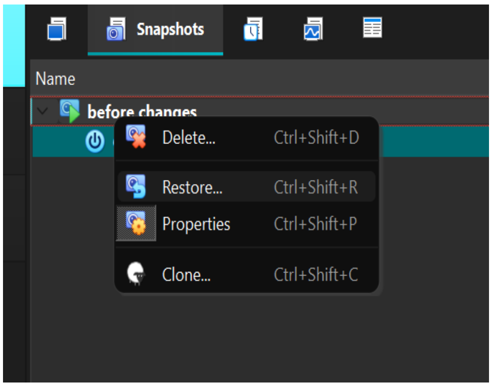
<- After deleting the file, a snapshot is created/saved ->

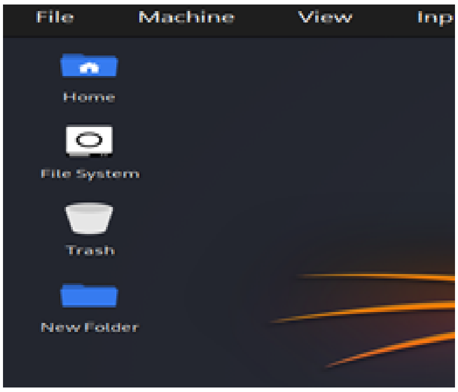


Restore Snapshot

1. Power off the Virtual Machine.
2. Go to the **Snapshots tab**
3. Select snapshot
4. Click **Restore**

After changing Restoring, the Virtual Machine to its previous state

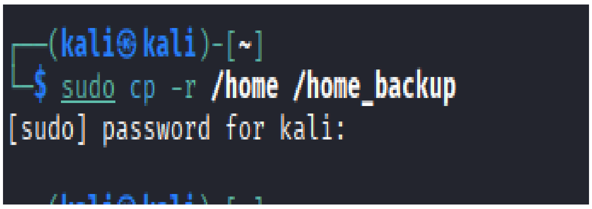




Alternative (Simple Backup Snapshot)

If you just want a quick backup:

sudo cp -r /home /home_backup



Important Tips

- Always take snapshots:
  - Before updates
  - Before installing tools
  - Before risky experiments (hacking tools, configs)
- Snapshots use **disk space**

Best Practice

Name snapshots like:

- clean-install
- before-tools
- after-config

VM Cloning and Export/Import

Step-by-Step Procedure:

Step 1: Take any existing VM

For the purposes of this lab, ensure you have a VM (for example, the Linux Mint VM from your previous tasks) that is fully shut down in VirtualBox's main window. It must not be in a "Saved" state.

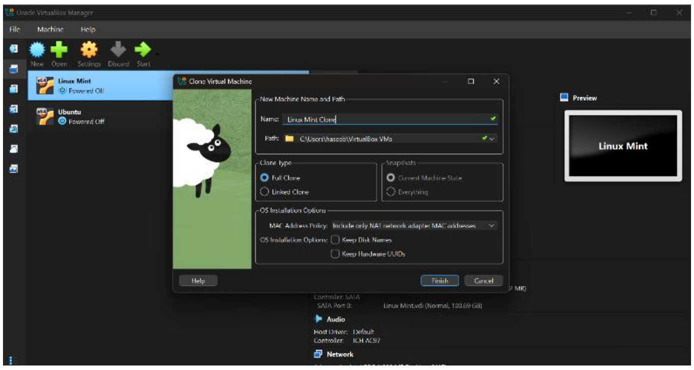
Step 2: Create a Full Clone

Cloning creates an independent, ready-to-run copy of a VM on the same host computer. A "Full Clone" creates a complete, separate copy of the VM's virtual disk, allowing it to operate independently of the source VM. It is ideal for creating a permanent, standalone duplicate, for example, for testing or development.

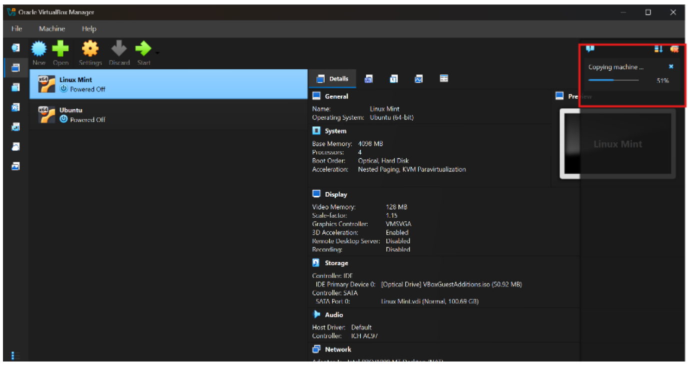
➢ Follow these steps to take your first screenshot ("Screenshot of cloning process"):

- In the VirtualBox Manager, right-click on your source VM and select "Clone".

- In the "Clone Virtual Machine" dialog box that opens, as shown in the example below, you will see several options:
- **Name:** Enter a name for your new VM (e.g., "Mint-Clone").
- **Path:** You can keep the default or choose a location with enough free space.
- **MAC Address Policy:** It's generally recommended to select "Generate new MAC addresses for all network adapters" to prevent network conflicts when both VMs are running.
- **Clone Type:** Select "Full clone".



Cloning VM:



- A new window titled "Clone Virtual Machine" in the Oracle VM VirtualBox Manager, showing the configuration options for creating a full clone of a virtual machine.
- Then, click "Clone" to start the process.

Step 3: Export VM as an OVA File

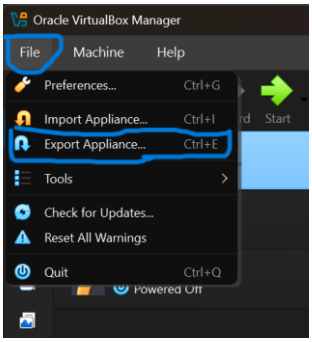
Exporting packages the entire VM (its configuration and disk images) into a single portable file, either an Open Virtualization Format Archive (OVA) file or a set of OVF files. This is the standard method for migrating a VM to a different virtualization platform or archiving it for later use.

What is an OVA?

An Open Virtualization Format Archive (OVA) is a single file (a .ova archive) containing all the data needed to recreate a virtual machine on another system, making migration simple.

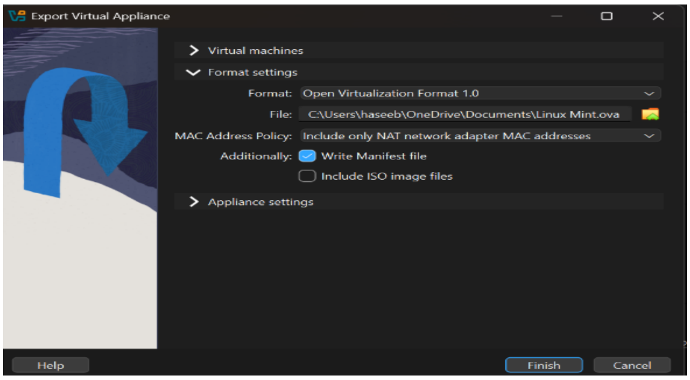
➢ Follow these steps to take your second screenshot ("Screenshot of exported file (.ova)"):

- Ensure your source VM is powered off.
- In the VirtualBox Manager menu bar, go to File → Export Appliance....
- Select the VM you want to export from the list and click Next.

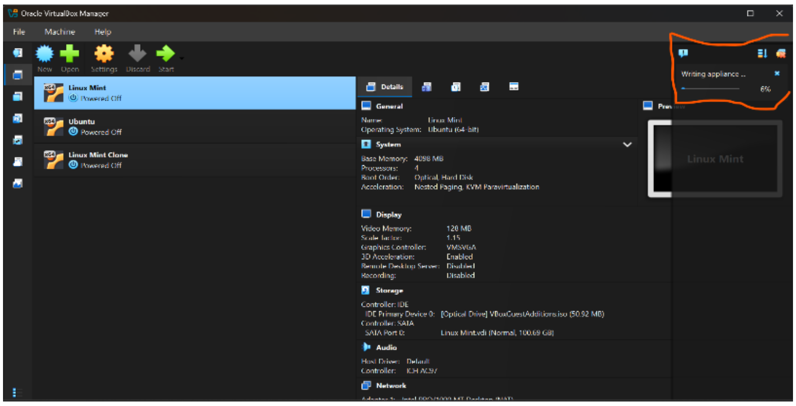


In the "Export Appliance Settings" window:

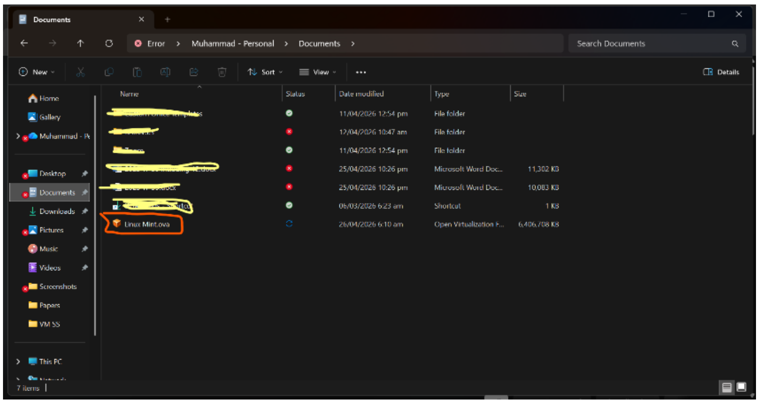
- **Format:** Choose "Open Virtualization Format Archive (OVA)" to create a single file.
- **File:** Select where you want to save the OVA file and give it a name (e.g., "MyVM.ova").
- Click Next.
- You can then review the VM settings. Leaving the defaults is generally fine.
- Click Finish to begin the export.



Exporting:



Once the export is complete, navigate to the location you saved the file. Check the .ova file in your file manager.

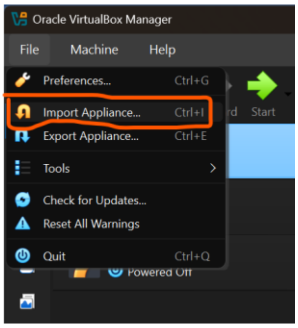


Step 4: Import the same OVA file back

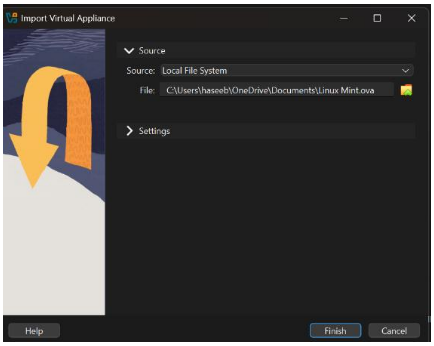
Importing is the process of taking an exported appliance file (an OVA file) and re-creating the virtual machine from it. This is how you can deploy a pre-configured VM on a new host.

➢ Follow these steps to take your third screenshot ("Screenshot of imported VM"):

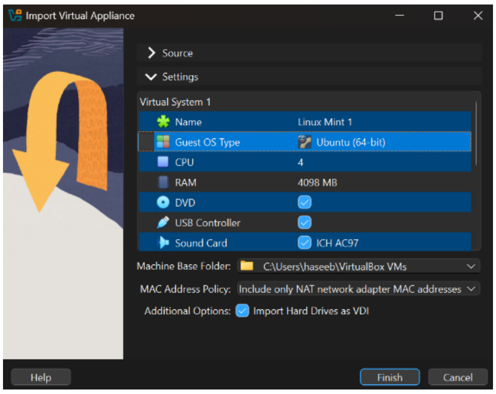
- In the VirtualBox Manager menu bar, go to File → Import Appliance....



- Click the folder icon and browse to select the .ova file you just exported.
- Click Next.



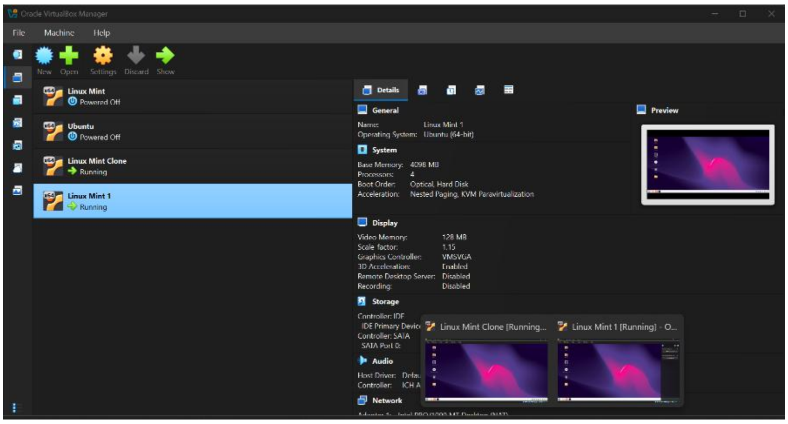
- In the "Settings" window, you can rename the VM to avoid confusion (for example, rename it to "Mint-Imported"). You can also change other settings like RAM, CPUs, or storage paths.



- Click Finish to start the import.

Once imported, you will see the new VM listed in the VirtualBox Manager.

Start both the clone (from Step 2) and the imported VM (using their new names) to check that they are working.



Explanation

Clone vs. Export/Import:

The main differences between these two operations are their purpose, output format, and where they are used:

Cloning a VM

- **Location:** Works on the same host computer.
- **Output:** It creates another native VirtualBox VM (a folder with .vbox and .vdi files) that is ready to run immediately.
- **Speed:** Generally very fast as it doesn't change the disk file format.
- **Use Case:** To quickly create another independent VirtualBox VM on the same machine for testing, development, or creating a backup copy.

Exporting and Importing a VM

- **Location:** Used to move a VM between different computers or to archive it.
- **Output:** It packages the VM into a standard, portable OVA (Open Virtualization Format Archive) file, which can be stored on a USB drive or uploaded to the cloud.
- **Disk Format Change:** During export, VirtualBox may convert the virtual disk to VMDK, the standard format used by VMware and other virtualization platforms.
- **Detail:** Because it re-packages the VM for portability, some specific hardware features may be reset or lost during the export/import process.
- **Use Case:** To distribute a VM to colleagues, migrate it to a data center, or create a long-term, hardware-independent backup.

LAB NO#09

Introduction To Shell & Shell 1st File

Introduction To Shell

What is Shell?

The shell is a program that takes your keyboard commands and passes them to the **Kernel** (the core of the Linux OS) to execute.

- **Terminal Emulator:** This is the window you open (usually "QTerminal" in Kali) to type your commands.
- **The Shell:** This is the actual "engine" running inside that window. In modern Kali Linux, the default shell is **ZSH** (Z Shell), though older versions used **Bash**.

ZSH vs. Bash

While **Bash** (Bourne Again Shell) is the industry standard for most Linux servers, Kali switched to **ZSH** as its default for several reasons:

- **Auto-completion:** You can press the Tab key to finish typing a command or file name.
- **Theming:** Kali's ZSH comes pre-configured with a "prompt" that shows your current directory and status in a clean, colorful way.
- **Plugin Support:** It handles advanced features like syntax highlighting (if you type a command correctly, it turns green; if not, it stays red).

Anatomy of the Kali Prompt

When you open the terminal in Kali, you see a line that looks like this: user@kali:~$

- **user:** Your current username.
- **kali:** The hostname (name of the computer).
- **~:** This symbol (the tilde) represents your **Home Directory** (/home/user).
- **$:** This indicates you are a **standard user**. If this changes to a **#**, it means you have **Root (Administrative)** privileges.

Core Concepts of the Kali Shell

- **Command Interpretation:** It receives commands, interprets them, and passes them to the operating system.
- **Command Line Interface (CLI):** Essential for penetration testers to control the target system efficiently.
- **Default Shell:** While Bash is the default, others like Sh or Csh exist; in modern Kali, Zsh is also commonly configured, but Bash commands are widely used.

Essential Shell Capabilities

- **File/Directory Management:** Using commands like ls (list), cd (change directory), mkdir (make directory), and rm (remove).
- **System Monitoring & Administration:** Managing user accounts and monitoring resources.
- **Automation (Shell Scripting):** Turning a series of commands into a .sh file to automate repetitive tasks.
- **Security Tools:** Direct access to powerful penetration testing tools pre-installed in Kali

Basic Shell Commands for Beginners

- **pwd**: Displays the current directory path.
- **ls -la**: Lists all files (including hidden) in detail.

- **grep**: Searches for specific patterns within files.
- **man <command>**: Shows the manual for a specific command.
- **history**: Displays previously executed commands.
- **chmod +x script.sh**: Gives executable permissions to a script.

Getting Started

1. **Open the Terminal:** The default terminal emulator (often QTerminal or GNOME Terminal) starts a new shell session.
2. **Verify Shell:** Type echo $SHELL to check which shell you are currently using.
3. **Basic Interaction:** Start with whoami to see your current user or ifconfig to check network interfaces.

- **Shebang Line:** All shell scripts should start with #!/bin/bash to ensure they run with the correct interpreter.
- **Networking:** Use netcat (nc -lvnp 2222) for initiating listeners and establishing reverse shells.

Shell 1st File (Basic.sh)

1. The Three Primary Modes

- **Command Mode:** The default mode when you open a file. Every keypress is interpreted as a command (e.g., "delete line," "save," or "move cursor"). You cannot type text here.
- **Insert Mode:** Used for typing text. You enter this by pressing i.
- **Last Line (Command-Line) Mode:** Used for saving, exiting, and advanced searching. You enter this by typing :.

2. Basic Navigation (Command Mode)

While you can use arrow keys in modern versions, the traditional vi keys allow you to navigate without moving your hands from the home row:

- **h** : Move left
- **j** : Move down
- **k** : Move up
- **l** : Move right
- **w** : Jump to the start of the next **word**
- **0 (zero)** : Jump to the **start** of the line
- **$** : Jump to the **end** of the line.

3. Editing Commands (Command Mode)

These commands help you manipulate text quickly:

- **i** : Enter **Insert** mode at the cursor.
- **x** : **Delete** the single character under the cursor.
- **dd** : **Delete** (cut) the entire current line.
- **yy** : **Yank** (copy) the current line.
- **p** : **Paste** the copied/deleted text after the cursor.

- **u** : **Undo** the last action.

4. Saving and Exiting (Last Line Mode)

To use these, press Esc to ensure you are in Command Mode, then type the colon : followed by the command:

- **:w : Write** (save) the file.
- **:q** : **Quit** (only works if there are no unsaved changes).
- **:wq** : **Write and Quit** (save and exit).
- **:q!** : **Quit without saving** (discard all changes).

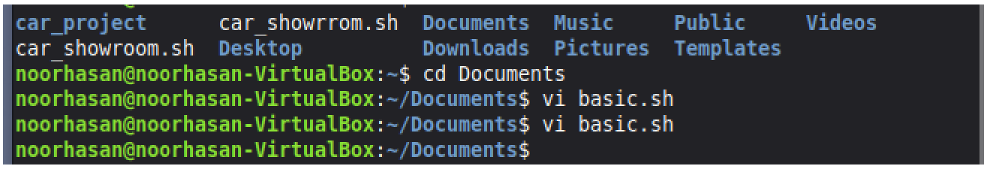
5. Searching

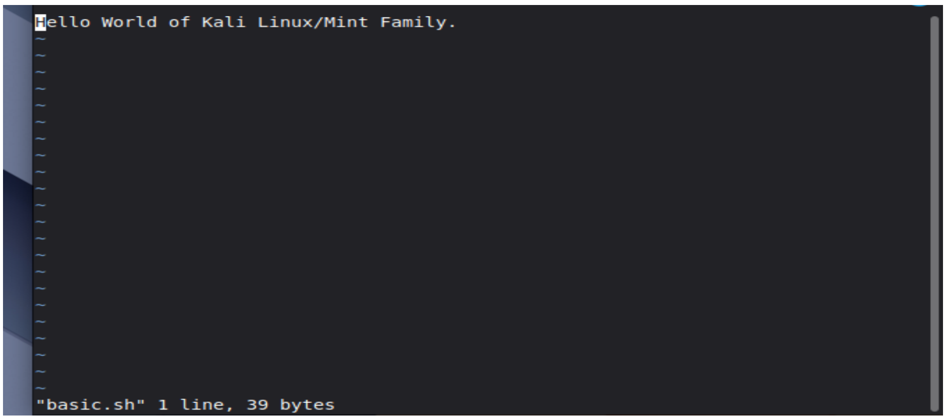
- **/word :** Search forward for "word". **Press n** to go to the next occurrence.
- **?word :** Search backward for "word".

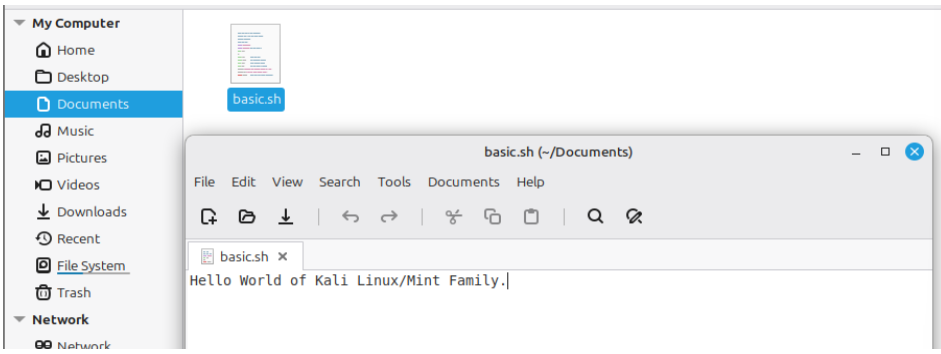
Typical Workflow Example

1. **Open file:** vi basic.sh
2. **Press i** to start typing your script.
3. **Press Esc** to stop typing and go back to Command Mode.
4. Move to a mistake **using h,j,k,l** and **press x** to delete it.
5. **Type :wq** and **press Enter** to save and get back to the terminal.

Creating Basic.sh File in Documents using







LAB NO#10

Shell 2nd File (Comment.sh) & 3rd File (Variable.sh)

Shell 2nd File (Comment.sh)

In Linux scripting, a **comment** is a line of text that the computer ignores during execution. Its sole purpose is to explain **what** the code is doing and **why** for the benefit of humans (yourself or your colleagues).

The Purpose of Comments

1. **Documentation:** Explains complex logic so don't forget how it works six months later.
2. **Debugging:** You can "comment out" a line of code (put a # in front of it) to temporarily disable it without deleting it.
3. **Metadata:** Often used at the top of a script to list the author, date, and version.

Create comment.sh using vi

1. Open the file: vi comment.sh
2. Press **i** to enter Insert Mode.

4. Press **Esc**, type **:wq**, and press **Enter**.

How Comments Work in Bash

- **The # Symbol:** Anything following a # on a line is ignored.
- **The Shebang (#!):** This is the **only** exception. Even though it starts with #, the ! tells the system this is a special instruction to find the command interpreter.
- **Multi-line Block:** Bash doesn't have a dedicated "official" multi-line comment like C++ (/* ... */), but using the : ' ' (colon and single quotes) trick effectively tells Bash to treat the block as a "do-nothing" string.

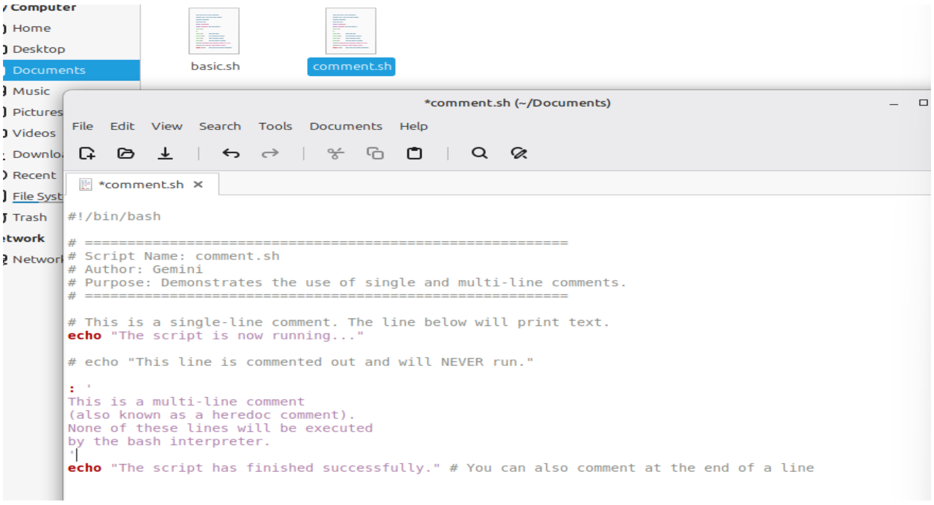
Creating a Comment.sh file.

```
noorhasan@noorhasan-VirtualBox:~/Documents$ vi basic.sh
noorhasan@noorhasan-VirtualBox:~/Documents$
noorhasan@noorhasan-VirtualBox:~/Documents$ vi comment.sh
noorhasan@noorhasan-VirtualBox:~/Documents$ vi comment.sh
```

Writing a Comment in shell.

```
#!/bin/bash

# ==========================================================
# Script Name: comment.sh
# Author: Gemini
# Purpose: Demonstrates the use of single and multi-line comments.
# ==========================================================

# This is a single-line comment. The line below will print text.
echo "The script is now running..."

# echo "This line is commented out and will NEVER run."

: '
This is a multi-line comment
(also known as a heredoc comment).
None of these lines will be executed
by the bash interpreter.
'
echo "The script has finished successfully." # You can also comment at the end of a line
~
~
```

Comment Text file in documents folder.



Shell 3rd File (Variable.sh)

In shell scripting, **variables** are containers that store data, such as text strings, numbers, or the output of system commands. Unlike languages like C++ or Java, Bash is "loosely typed," meaning you don't need to declare if a variable is an integer or a string.

Key Rules for Shell Variables

1. **No Spaces:** You must not have spaces around the = sign.
   - NAME="User" is **Correct**.
   - NAME = "User" is **Wrong**.
2. **Case Sensitivity:** NAME and name are two different variables.
3. **Accessing Variables:** Use the **$** symbol to retrieve the value stored in a variable.

Create variable.sh using vi

1. Open the file: vi variable.sh
2. Press **i** to enter Insert Mode.
3. Press **Esc**, type **:wq**, and press **Enter**.

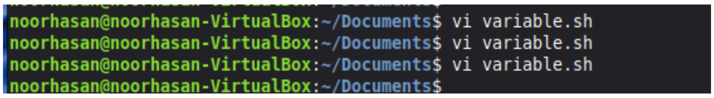
Types of Variables in Linux

- **Local Variables:** Defined within the script and only available while the script is running.
- **Environment Variables:** Variables available system-wide (e.g., $PATH, $HOME, $SHELL). You can see all of these by typing the env command in your terminal.

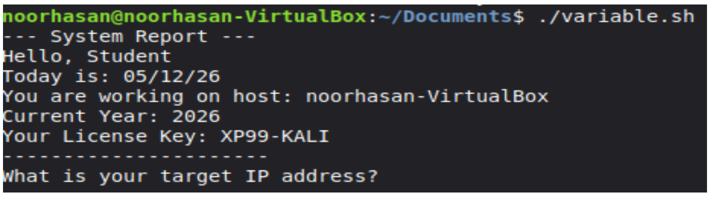
Special Variables:

- **$0:** The name of the script itself.
- **$1 to $9:** The first nine arguments passed to the script from the command line.
- **$#:** The number of arguments passed to the script.
- **$?:** The exit status of the last command (0 means success).

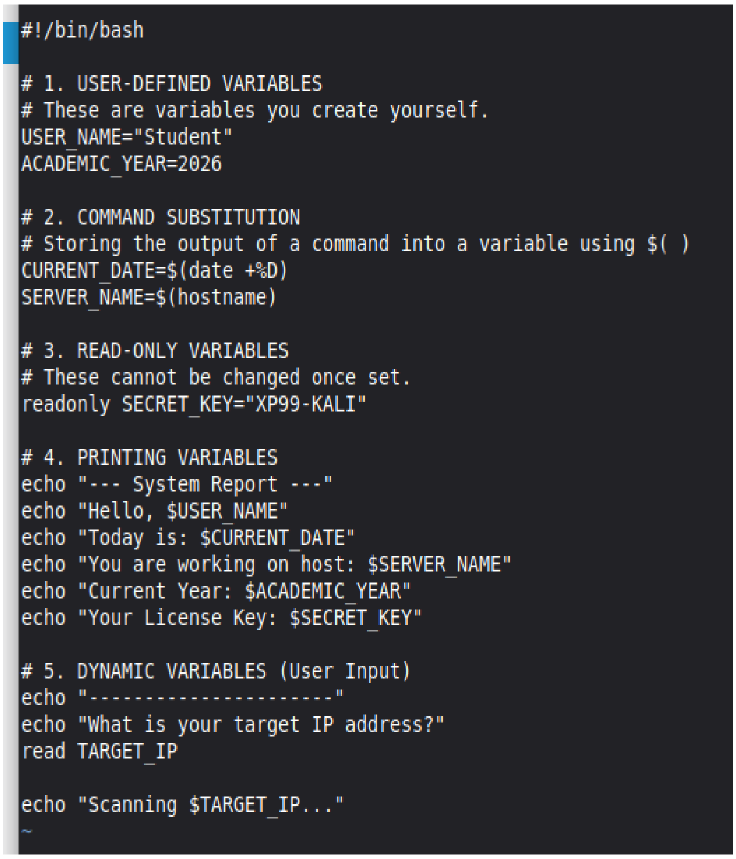
Creating a Variable.sh file.



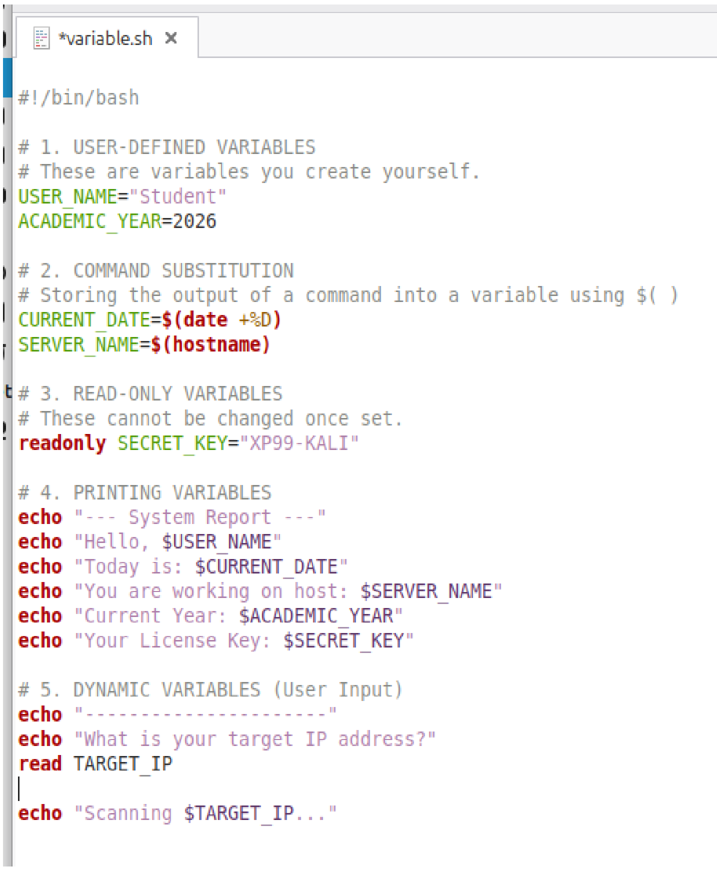
Displaying the Variable.sh output in Terminal.



Writing Variable code in the shell



Displaying Output in Text Editor



LAB NO#11

Shell 4th File (Constant-Variable.sh) & 5th File (Array.sh)

Shell 4th File (Constant-Variable.sh)

In shell scripting, a **constant** is a variable whose value is intended to remain unchanged throughout the execution of the script. In Bash, you enforce this by using the readonly command.

Purpose of Constant Variables

- **Prevent Accidental Overwrites:** They protect critical data (like API keys, configuration paths, or mathematical constants) from being changed by other parts of the script.

- **Security:** Using readonly ensures that sensitive strings, like a SECRET_KEY, cannot be manipulated while the script is active.
- **Readability:** Traditionally, constants are written in **UPPERCASE** to distinguish them from regular variables.

Create constant_variable.sh using vi

1. Open the file: vi constant_variable.sh
2. Press **i** to enter Insert Mode.
3. Press **Esc**, type **:wq**, and press **Enter**.

What to expect in the terminal:

When the script reaches the line where you try to change **COMPANY_NAME**, the shell will display an error message similar to: **line 18: COMPANY_NAME: readonly variable.**

Creating a Constant_Variable.sh file.

```
noorhasan@noorhasan-VirtualBox:~/Documents$ vi constant_variable.sh
noorhasan@noorhasan-VirtualBox:~/Documents$ ./constant_variable.sh
```

Writing the constant_variable code in file.

```
#!/bin/bash

# --- 1. DEFINING CONSTANTS ---
# Use the 'readonly' command to make a variable a constant
readonly PI=3.14159
readonly COMPANY_NAME="TechCorp Solutions"
readonly LOG_FILE="/var/log/system_check.log"

# --- 2. USING CONSTANTS ---
echo "Welcome to $COMPANY_NAME"
echo "The value of PI is approximately: $PI"
echo "All events will be logged to: $LOG_FILE"

# --- 3. ATTEMPTING TO CHANGE A CONSTANT ---
echo "----------------------------------------"
echo "Attempting to change the COMPANY_NAME..."

# This next line will trigger an error in the terminal
COMPANY_NAME="HackerCorp"

echo "The script continues... Company is still: $COMPANY_NAME"
```

Running the constant_variable.sh code in terminal.

```
noorhasan@noorhasan-VirtualBox:~/Documents$ vi constant_variable.sh
noorhasan@noorhasan-VirtualBox:~/Documents$ ./constant_variable.sh
Welcome to TechCorp Solutions
The value of PI is approximately: 3.14159
All events will be logged to: /var/log/system_check.log
----------------------------------------
Attempting to change the COMPANY_NAME...
./constant_variable.sh: line 20: COMPANY_NAME: readonly variable
The script continues... Company is still: TechCorp Solutions
```

Shell 5th File (Array.sh)

In shell scripting, an **array** is a variable that can hold multiple values under a single name. Instead of creating user1, user2, and user3, you can store them all in a users array.

Key Concepts of Shell Arrays

- **Indexing:** Arrays in Bash are **zero-based**. The first element is at index 0.
- **Syntax:** Use parentheses () to define the array and square brackets [] to access specific elements.
- **The @ Symbol:** Using [@] allows you to refer to all elements in the array at once.

Create array.sh using vi

1. Open the file: vi array.sh
2. Press **i** to enter Insert Mode.
3. Press Esc, type :wq, and press Enter.

How to Run and View the Script

To see this in action, follow the permission steps we discussed earlier:

Set permissions: **chmod 755 array.sh**

View the code to double check: **cat array.sh**

Execute the script: **./array.sh**

Pro-Tip: Array Slicing

If you want to print only a specific range of an array (for example, the first two items), you can use the slicing syntax: echo ${TOOLS[@]:0:2} This tells Bash: "Start at index 0 and show me 2 elements."

Creating an Array.sh file.

```
noorhasan@noorhasan-VirtualBox:~/Documents$
noorhasan@noorhasan-VirtualBox:~/Documents$ vi array.sh
noorhasan@noorhasan-VirtualBox:~/Documents$ cat array.sh
```

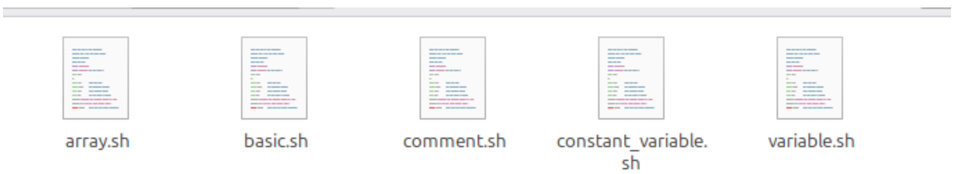
Writing code of array.sh and checking using cat in file.

```
#!/bin/bash

# --- 1. DEFINING AN ARRAY ---
# Elements are separated by spaces
TOOLS=("Nmap" "Metasploit" "Wireshark" "BurpSuite")

# --- 2. ACCESSING ELEMENTS ---
echo "The first tool is: ${TOOLS[0]}"
echo "The second tool is: ${TOOLS[1]}"

# --- 3. PRINTING ALL ELEMENTS ---
echo "All tools in the hacking kit: ${TOOLS[@]}"

# --- 4. ARRAY LENGTH ---
echo "Total number of tools: ${#TOOLS[@]}"

# --- 5. ADDING/UPDATING ELEMENTS ---
TOOLS[4]="Hydra"              # Adding a 5th element
TOOLS[0]="ZAP"                 # Overwriting the 1st element

echo "Updated first tool: ${TOOLS[0]}"
echo "New total count: ${#TOOLS[@]}"

# --- 6. LOOPING THROUGH AN ARRAY ---
echo "--------------------------"
echo "Listing all tools via loop:"
for tool in "${TOOLS[@]}"; do
    echo "Tool Name: $tool"
done
```

Running the array.sh code in terminal.

```
noorhasan@noorhasan-VirtualBox:~/Documents$ chmod 755  array.sh
noorhasan@noorhasan-VirtualBox:~/Documents$ ./array.sh
The first tool is: Nmap
The second tool is: Metasploit
All tools in the hacking kit: Nmap Metasploit Wireshark BurpSuite
Total number of tools: 4
Updated first tool: ZAP
New total count: 5
--------------------------
Listing all tools via loop:
Tool Name: ZAP
Tool Name: Metasploit
Tool Name: Wireshark
Tool Name: BurpSuite
Tool Name: Hydra
```

All The Shell Files



LAB NO#12

Installing & Configuring DHCP, DNS, FTP Server

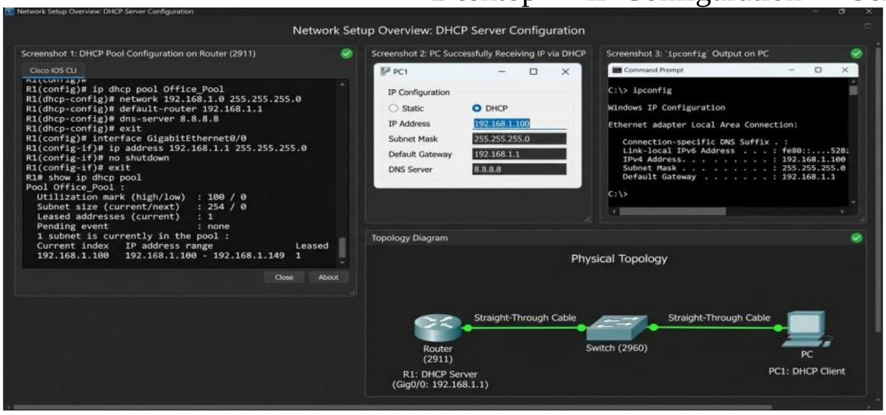
DHCP Server Configuration Requirements:

- 1 × Router (2911)
- 1 × Switch (2960)
- 1 × PC
- Straight-Through Cables

Procedure:

1. **Build the topology: PC → Switch → Router**
2. **Configure the router as a DHCP Server:**
   - Go to Router → **Config** → **DHCP**
   - Create new Pool:
     - Pool Name: Office_Pool
     - Network: 192.168.1.0
     - Subnet Mask: 255.255.255.0
     - Default Gateway: 192.168.1.1
     - DNS Server: 8.8.8.8
     - Start IP: 192.168.1.100
     - Maximum Users: 50
3. **Configure Router Interface:**
   - GigabitEthernet0/0 → IP Address: 192.168.1.1
   - Subnet Mask: 255.255.255.0
   - Port Status: **On**
4. **On PC:**

   Desktop → IP Configuration → Select **DHCP**



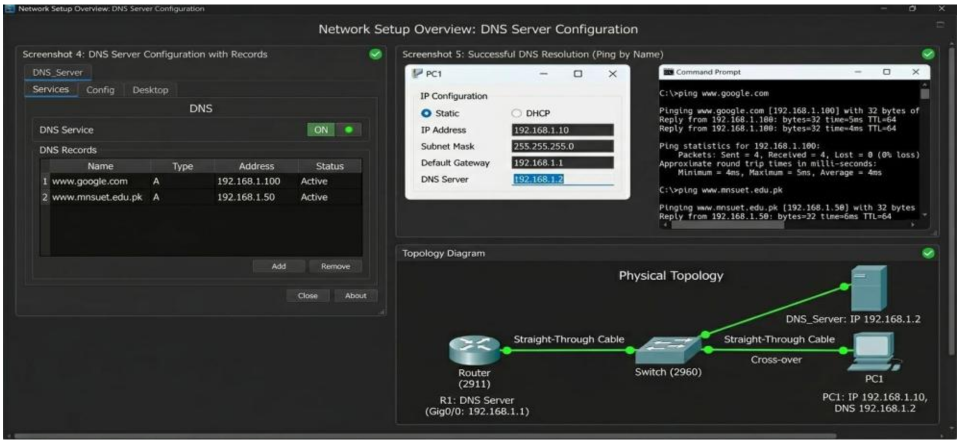
DNS Server Configuration Requirements:

- 1 × Router
- 1 × Switch
- 1 × Server
- 1 × PC

Procedure:

1. **Assign a static IP to the server:**
   - Server IP: 192.168.1.2
   - Gateway: 192.168.1.1
2. **Enable DNS Service:**
   - Server → **Services** → **DNS** → Turn **ON**
   - Add DNS Records:
     - Name: www.google.com → Address: 192.168.1.100

      ▪ Name: www.mnsuet.edu.pk → Address: 192.168.1.50
3. **Configure PC:**
   - IP: DHCP or Static (192.168.1.10)
   - DNS Server: 192.168.1.2
4. **Test:**
   - Command Prompt → ping www.google.com

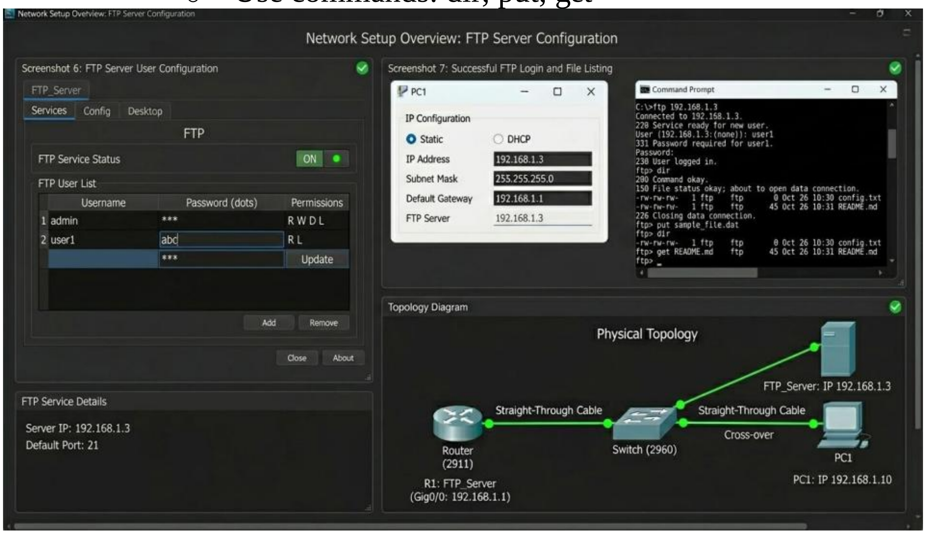


FTP Server Configuration Requirements:

- 1 × Router
- 1 × Switch
- 1 × Server
- 1 × PC

Procedure:

1. Assign IP to FTP Server:
   - Server IP: 192.168.1.3
   - Gateway: 192.168.1.1
2. Enable FTP Service:
   - Server → **Services** → **FTP** → Turn **ON**
   - Create Users:
      ▪ Username: admin → Password: 123
      ▪ Username: user1 → Password: abc
3. On PC:
   - Command Prompt → ftp 192.168.1.3
   - Login with username and password
   - Use commands: dir, put, get



LAB NO#13

Implementing Server Backup Strategies & Restoring Data

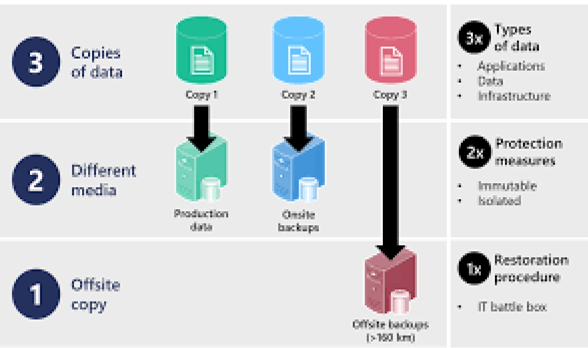
Implementing Server Backup Strategies

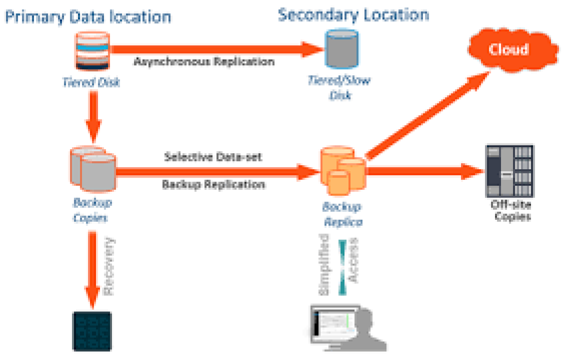
What are Server Backup Strategies?

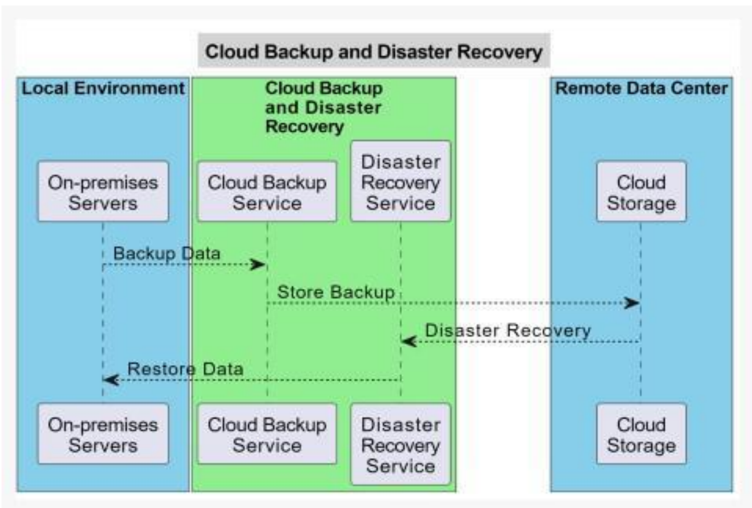
A server backup strategy is a defined plan for creating duplicate copies of data to ensure its availability after a disaster, accidental deletion, or cyberattack. A successful strategy balances backup speed, storage costs, and recovery time by combining different technical methods with organizational policies like the **3-2-1 rule**.

Core Backup Strategies & Methods

- **3-2-1 Backup Rule:** The standard for data protection: 3 total copies of data, 2 different media types (e.g., local NAS + cloud), 1 copy stored offsite.
- **Backup Types:**
  - **Full Back up:** Complete copy, simplest to restore but slowest and highest storage cost.
  - **Incremental Backup:** Captures only data changed since the last backup (any type), saving time and space.
  - **Differential Backup:** Captures data changed since the last *full* backup.
- **Hybrid Approach:** Combining local storage (high-speed restoration) with cloud storage (offsite protection) is ideal.







Implementation Steps

1. **Identify Data and Set Policies:** Identify critical data and define the **Recovery Point Objective (RPO)**—how much data you can afford to lose—and **Recovery Time Objective (RTO)**—how fast you need to recover.
2. **Automate Schedules:** Use tools to schedule automated backups to run during off-hours, ensuring consistency.
3. **Choose Storage Destinations:** Use a mix of external hard drives, NAS (Network Attached Storage), and reputable cloud services like Amazon AWS.
4. **Encrypt and Secure:** Utilize AES-256 encryption for stored data and manage keys separately.
5. **Enable Immutable Backups:** Utilize storage that prohibits modifications for a set period (e.g., 30+ days) to protect against ransomware.

Best Practices for Maintenance

- **Test Restores Regularly:** Periodically test the restoration process (monthly or quarterly) to verify that backups are functional.
- **Monitor Backups:** Review daily logs to address failed backups immediately.
- **Retention Management:** Define clear retention policies for how long data is stored to meet compliance requirements.

Restoring Data in Server Backup Strategies

Restoring server data involves using backed-up copies—full, incremental, or differential-to restore systems via methods like in-place restoration, volume recovery, or fast streaming recovery. Best practices require the 3-2-1 rule (3 copies, 2 media, 1 offsite) and regular testing to ensure data integrity.

Key Data Restoration Strategies

- **Full System Restore:** Replacing all lost data with the most recent full-image backup. This is the fastest, most straightforward method but can be time-consuming to create.
- **Incremental Restore:** Requires the last full backup plus each subsequent incremental backup, offering faster backup times but slower restoration.
- **Differential Restore:** Restores the last full backup followed only by the latest differential backup (changes since the last full backup), offering faster restoration than incremental.
- **Streaming/Instant Recovery:** Instantiates virtual machine (VM) volumes instantly on production storage, streaming data from the backup, ideal for minimizing downtime.
- **Volume/VMDK Recovery:** Restores entire Virtual Machine Disks (VMDK) quickly, often used when multiple VMs fail simultaneously.
- **Binary/Database Restore:** Specialized SQL database restoration using binary copies, which can be done offline or online via commands like **dsconf restore.**

Best Practices for Effective Restoration

- **3 2 1 1 0 Rule:** Maintain **3 copies** of data, on **2 different media** types, with **1 off-site** copy, **1 immutable/air-gapped** copy, and **0 errors verified** by testing.
- **Regular Restoration Testing:** Regularly restore to a controlled environment to verify backup integrity, ensuring data is not corrupted and recovery times are met.
- **Automate Backups:** Automate daily or weekly schedules to eliminate human error, ensuring full backups are available when needed.

- **Define Objectives:** Set strict Recovery Time Objectives (RTO—how fast to recover) and Recovery Point Objectives (RPO—how much data loss is acceptable).

Common Pitfalls to Avoid

- **Failure to Test:** 50% of restores fail, usually due to incomplete backups or lack of testing.
- **Ignoring Retention Policies:** Without a clear policy, old backups can become a liability or hinder recovery.
- **Lack of Encryption:** Unencrypted backups pose a security risk during the restoration process.

LAB NO#14

Intro to Monitoring Tools to Detect Network Breaches

Introduction to Monitoring Tools

Monitoring tools are software applications that track the health, performance, and availability of IT infrastructure, networks, and applications in real-time. They provide visibility into system metrics—like CPU usage or network traffic—and alert administrators to potential issues, helping ensure operational efficiency and prevent downtime.

Core Purpose of Monitoring

The primary goal is to ensure **observability**—the ability to understand the internal state of a system by looking at the data it produces.

- **Proactive Detection:** Identifying issues (like a CPU spike or low disk space) before they cause a system crash.
- **Performance Analysis:** Tracking how fast an application responds to users.
- **Uptime Verification:** Confirming that services (like your DHCP or DNS servers) are running and accessible.
- **Historical Trending:** Looking at data over months to predict when you will need to buy more storage or upgrade hardware.

Key Aspects of Monitoring Tools

- **Types:** Tools range from network monitoring (routers, switches) to Application Performance Monitoring (APM) (server response times, user experience).
- **Functionality:** They gather data via agents or agentless technology, providing centralized dashboards for visualization, data analysis, and proactive alerting.
- **Key Techniques:** Key techniques include log analysis (e.g., ELK Stack), network traffic analysis, and synthetic monitoring.
- **Benefits:** These tools optimize performance, reduce downtimes, and aid in capacity planning.

Popular Monitoring Solutions

- **Prometheus:** Often used for cloud-native metrics and Kubernetes monitoring, featuring a unique "pull" model.
- **Datadog:** Known for comprehensive, unified observability across environments.

- **Dynatrace:** An AI-powered APM tool.
- **Nagios:** Used for monitoring system and application health.

How Monitoring Works (The Workflow)

1. **Data Collection:** Tools use "Agents" (small software installed on the server) or protocols like **SNMP** (Simple Network Management Protocol) to gather data.
2. **Threshold Setting:** You define what is "normal." For example, "Alert me if CPU usage stays above 90% for more than 5 minutes."
3. **Visualization:** Data is displayed on **Dashboards** (graphs, charts, and "green/red" lights) for easy viewing.
4. **Alerting:** When a threshold is broken, the tool sends a notification via email, SMS, or Slack to the administrator.

Why Use Monitoring Tools?

They are crucial for identifying bottlenecks, such as slow database queries, which can directly affect user experience. They also help ensure that system metrics stay within defined, acceptable limits, alerting teams before problems escalate.

Detection of Network Breach

Network breach detection involves using monitoring tools to identify malicious activities such as data exfiltration or lateral movement through real-time traffic analysis, anomaly detection, and signature-based identification. Common tools include Splunk, Suricata, Wireshark, and Zeek, which track metrics like latency, bandwidth spikes, and unauthorized configuration changes.

Key Types of Monitoring Tools for Detection

- **Intrusion Detection/Prevention Systems (IDS/IPS):** Tools like **Suricata** and **Snort** analyze traffic in real-time, matching packets against known attack signatures and behavioral patterns to stop threats.
- **Security Information and Event Management (SIEM):** Solutions such as **Splunk** ingest data from servers, apps, and cloud services to provide visibility into network activities and generate actionable alerts.
- **Network Packet Analyzers: Wireshark** is used for deep, real-time packet inspection to identify security threats, while **TCPdump** serves as a command-line alternative.
- **Network Performance & Traffic Monitoring:** Tools like **Nagios**, **SolarWinds**, and **Datadog** monitor bandwidth usage and identify unusual traffic patterns, such as unexpected outbound connections.
- **Configuration Monitoring:** Tracks changes to network device settings to identify potential security lapses or unauthorized changes.

Indicators of a Breach to Monitor

- **Unusual Outbound Traffic:** Large data transfers (data exfiltration) or spikes in traffic volume.
- **Lateral Movement:** Unusual connections between internal systems, often detected by IDS/IPS.
- **Unauthorized Access:** Changes in access controls or unexpected, successful logins.
- **Behavioral Anomalies:** Deviations from standard network usage patterns.

Top Tools for 2026

- **Splunk:** For deep visibility, though often costly.
- **Suricata/Snort:** High-performance threat detection.
- **Zeek:** Detailed, long-term traffic analysis.
- **Elastic ELK Stack:** For analyzing large, diverse data sets.
- **Cisco Stealthwatch:** Known for network threat visibility.
- **Auvik/LogicMonitor:** Often used by MSPs for, for example, cloud-first and on-premises environments.

Effective monitoring also involves using tools like **WMI** (Windows Management Instrumentation) and **SNMP** (Simple Network Management Protocol) to gather data on the health of network components.

LAB NO#15

Intro & use of Ping, Traceroute, and Wireshark to analyze Traffic

Intro of Ping, Trace-Route, and Wireshark

Ping, Traceroute, and Wireshark are essential, foundational network diagnostic tools used to test connectivity, map paths, and analyze traffic, respectively. **Ping checks** if a host is reachable, **Traceroute maps** the route path, and **Wireshark provides** in-depth, packet-level analysis.

1. Ping (Packet InterNet Groper)

The Purpose: To test if a specific host (website or IP address) is reachable and to measure how long it takes for data to be returned.

- **How it works:** It sends a small packet called an **ICMP Echo Request**. If the destination is "alive," it sends back an **ICMP Echo Reply**.
- **Key Metric: Latency (ms)** the time it takes for the round trip. A "Time Out" means the destination is down or blocking your request.
- **Usage:** ping 8.8.8.8 or ping google.com.

2. Traceroute (tracert on Windows)

The Purpose: To show the exact path a packet takes from your computer to a destination, listing every "hop" (router) along the way.

- **How it works:** It reveals the IP addresses of all the routers between you and the target. This is useful for finding exactly *where* a connection is failing.
- **Analogy:** If Ping tells you that a package didn't arrive, Traceroute shows you exactly which post office lost it.
- **Usage:** traceroute google.com (Linux/Mac) or tracert google.com (Windows).

3. Wireshark

The Purpose: To capture and interactively browse the traffic running on a computer network. It is known as a **Network Protocol Analyzer**.

- **How it works:** It intercepts "packets" as they travel through your network interface card (NIC) and breaks them down into a readable format. You can see the actual content of the data (unless it is encrypted).
- **In Security:** It is used to detect a network breach, find "cleartext" passwords in unencrypted protocols (like **FTP** or **HTTP**), and troubleshoot complex application errors.
- **The View:** It provides a "Packet List" (summary), "Packet Details" (layers of the OSI model), and "Packet Bytes" (the raw hex/data).

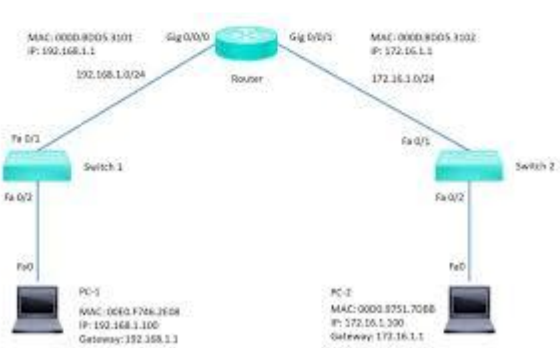
Use of Ping, Trace-Route, and Wireshark to Analyze Traffic

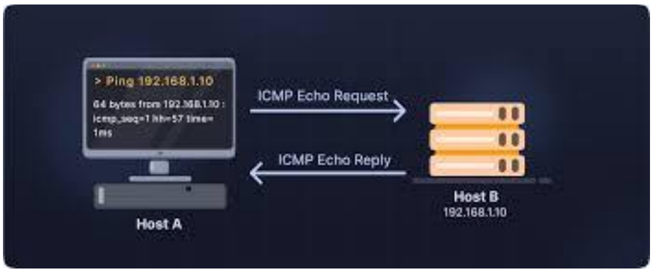
Ping and Traceroute are command-line tools for checking connectivity, latency, and pathing, while Wireshark provides deep packet inspection (DPI) to visualize the actual traffic exchange, making them ideal for diagnosing network bottlenecks, packet loss, and routing issues.

1. Ping: Testing Connectivity (ICMP)

Ping uses Internet Control Message Protocol (ICMP) Echo Request/Reply messages to test reachability and calculate round-trip time (RTT).

- **Usage:** ping <destination> (e.g., ping google.com).
- **Wireshark Analysis:**
  - Filter by icmp.
  - **Echo Request (Type 8):** Packet going to the target.
  - **Echo Reply (Type 0):** Packet coming back from the target.
  - **Analysis:** Check for missing replies (packet loss) or consistently high RTT values.



2. Traceroute: Mapping the Network Path

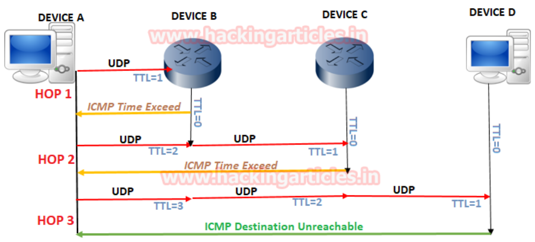
Traceroute maps the path to a destination, listing intermediate routers (hops) and identifying where bottlenecks occur by increasing the Time-To-Live (TTL) field in packet headers. [1, 2]

- **Usage:** traceroute <destination> (Linux/macOS) or tracert <destination> (Windows).

Wireshark Analysis:

- Filter by icmp or udp (depending on the OS implementation).
- **Time-to-live exceeded (ICMP Type 11):** These are sent by routers when the TTL reaches 0.
- **Analyze:** Inspect the source IP of "Time Exceeded" messages to map every step of the journey.

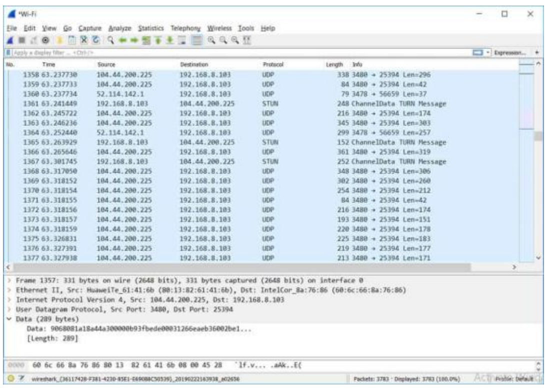
Working of Traceroute



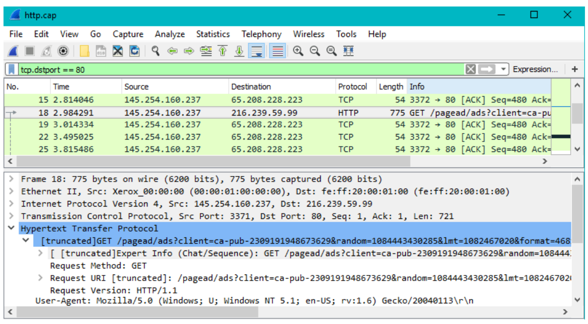
3. Wireshark: Deep Packet Inspection

Wireshark captures live packets, allowing for detailed analysis of the traffic generated by Ping and Traceroute.

- **Capture Strategy:**
  1. Open Wireshark and select your interface.
  2. Start capturing and running ping or traceroute in the terminal.
  3. Apply display filters: icmp or ip.addr == <target_ip>.
- **Key Inspection Metrics:**
  - **Protocol Details:** Examine Ethernet, IPv4, and ICMP headers.
  - **TTL Tracking:** Watch how TTL changes with each hop in a traceroute.
  - **Latency Identification:** Pinpoint which specific router in the path is causing latency spikes.
  - **Security Check:** Look for abnormally high frequency of ICMP traffic, which could indicate a DoS attack.



Wireshark is using a filter for Traffic.



Common Scenarios

- **Slow Connectivity:** Use **Traceroute** to identify which hop has a high average RTT, then use **Wireshark** to inspect that packet's behavior.
- **Host Unreachable:** Use **Ping** to confirm. If it fails, use **Wireshark** to check for Destination Unreachable (ICMP Type 3) messages.
- **Intermittent Connection:** Run a persistent **Ping** (-t) while capturing in **Wireshark** to catch specific packets that are dropped.

LAB NO#16

Explain the Use of the Following Tools

These five tools represent the backbone of modern infrastructure, ranging from running a single virtual computer on your laptop to managing massive global data centers using code.

VM-Ware

VMware (Virtualization)

The Purpose: VMware is a virtualization platform that allows you to run multiple "Virtual Machines" (VMs) on a single physical computer.

Use: Virtualization and Private Cloud Infrastructure

- **How it works:** It uses a "Hypervisor" to slice up your physical hardware (CPU, RAM, Disk) and share it among different operating systems. You could run Kali Linux, Windows Server, and Ubuntu at the same time on one laptop.
- **Primary Use:** Testing software in different environments, setting up home labs for networking practice, and consolidating server hardware in data centers.
- **Benefits:** Reduces hardware costs, improves server utilization, and increases IT agility.

Net-Sim

NetSim (Network Simulation)

The Purpose: NetSim is used to design, simulate, and analyze the behavior of computer networks without needing expensive physical hardware like routers and switches.

Use: Network Simulation and Performance Analysis

- **How it works:** It provides a digital environment to drag and drop network devices, connect them, and test protocols like OSPF, BGP, or 5G/6G signals.
- **Primary Use:** Network R&D, performance modeling, and educational training for certifications (like CCNA/CCNP). It is much deeper than a simple "drawing" tool because it calculates actual packet math.
- **Benefits:** Enables testing of network behaviors without expensive physical hardware prototypes, accelerating R&D.

AWS

AWS (Amazon Web Services)

The Purpose: AWS is the world's most comprehensive cloud platform, offering over 200 fully featured services from data centers globally.

Use: Public Cloud Computing and Services

- **How it works:** Instead of buying your own servers, you "rent" computing power, storage, and databases from Amazon. You only pay for what you use (on demand).
- **Primary Use:** Hosting websites, storing massive amounts of data (S3), running applications (EC2), and deploying AI/Machine Learning models.
- **Benefits:** Scalability, flexibility, high availability, and pay-as-you-go pricing.

Cloud-Sim

CloudSim (Cloud Research Simulation)

The Purpose: Unlike AWS (which is a real cloud), CloudSim is a **research toolkit** for simulating cloud computing environments.

Use: Modeling and Simulation of Cloud Environments

- **How it works:** It is a Java-based framework used by researchers to model "What if" scenarios. For example: "What happens to energy consumption if I move 1,000 tasks between two data centers?"
- **Primary Use:** Academic research in cloud scheduling, energy efficiency, and resource provisioning. It allows you to test cloud theories for free before spending money on a real provider like AWS.
- **Benefits:** Free to use, allows simulation of large-scale environments on a single machine, and removes the risk associated with testing on real cloud infrastructure.

Terra-Form

Terraform (Infrastructure as Code - IaC)

The Purpose: Terraform allows you to build, change, and version your infrastructure safely and efficiently using **configuration files** instead of clicking buttons in a dashboard.

Use: Infrastructure as Code (IaC) and Provisioning

- **How it works:** You write a script (using HCL - HashiCorp Configuration Language) that says, "I want 3 web servers and 1 database in AWS." When you run it, Terraform automatically talks to AWS and builds them for you.
- **Primary Use:** Automation. If you need to set up 100 identical servers, you don't do it manually; you use Terraform to deploy them in seconds.
- **Benefits:** Eliminates manual GUI clicking for provisioning, reduces errors, and allows infrastructure changes to be reviewed.